

- I appreciate the nice typing & neatness, but you can write portions of the report and scan them in, especially when calculations are involved. $\rightarrow$ Terms are not well defined.
- W_2 & W_3 are incorrect. Egn is right, but numbers are wrong.
- Illustrations are needed to explain calculations. You might have a fellow student read through this to see if they can understand it.
- When calculating interference, $b_{min} = .0006''$.

Project 3: Clock Tower Geartrain

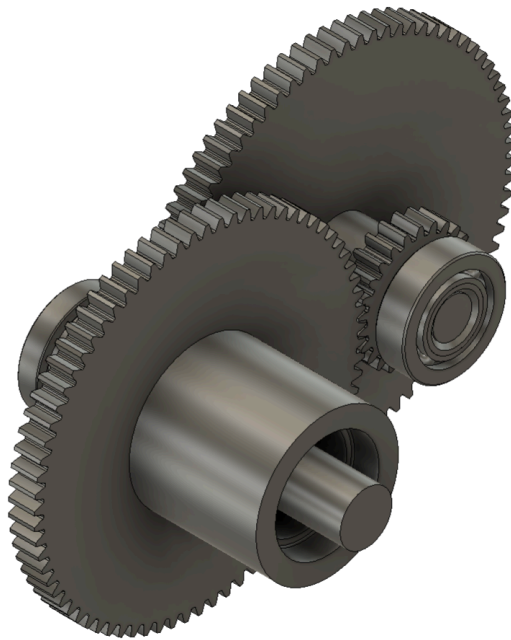
Machine Elements and Fatigue Design MAE 3620

April 1st, 2026

Anthony Bartolotti, Ty Davin, Matthew Herring, Patrick Jacobs

Good job.

90



- FBD's don't include any torques?
- Expression for max torque due to interference is wrong. What is μ ? .3, .5 or .8?

Executive Summary

Gear Train

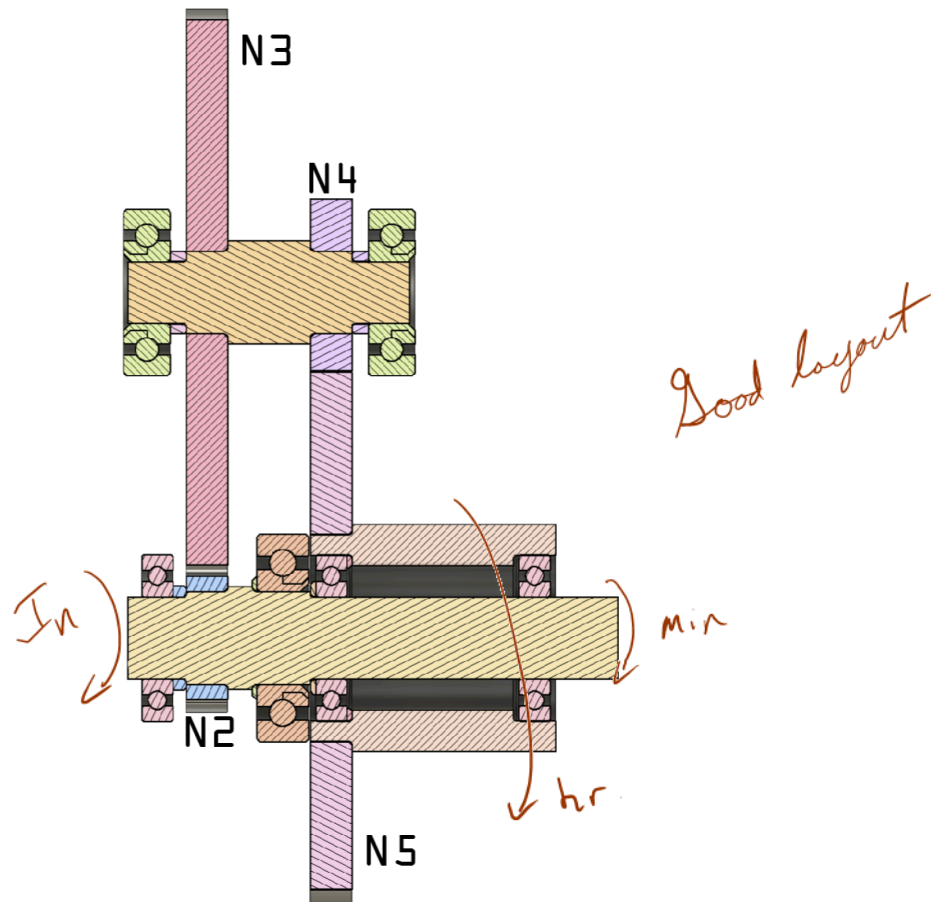


Figure 1. Cutaway Diagram

Gear Information

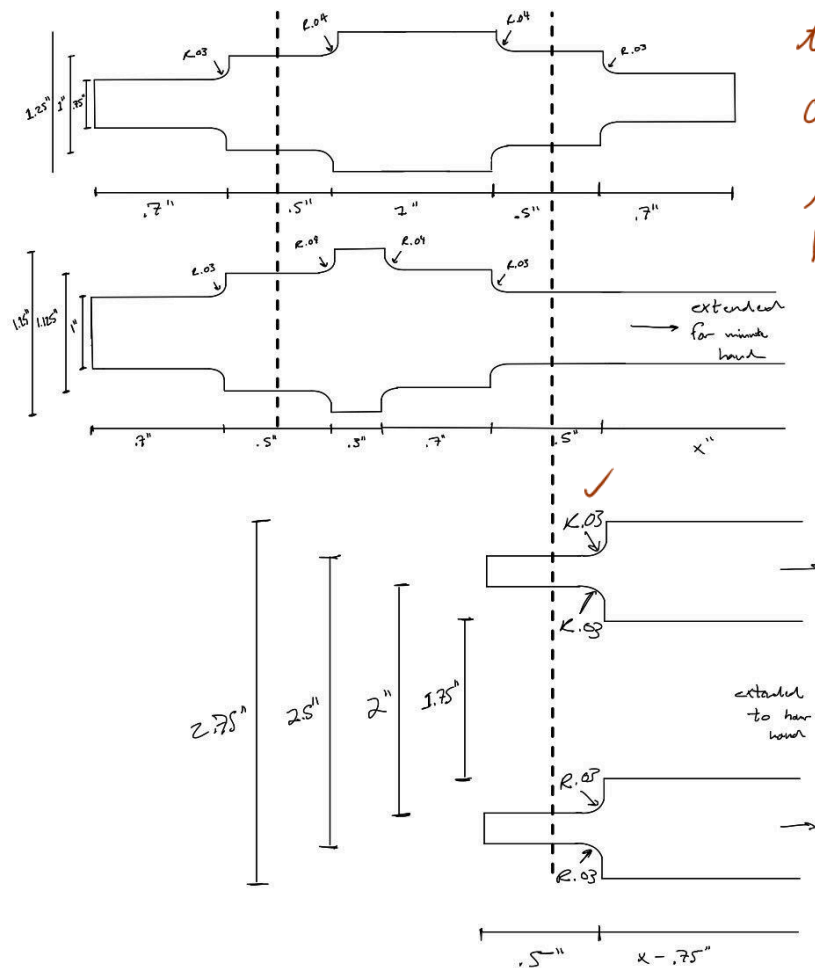
Gear #	Pitch (°)	# of Teeth	Face Width (in.)
N2	20	20	0.5
N3	20	80	0.5
N4	20	25	0.5
N5	20	75	0.5

Do you mean pressure angle?
In drawings, $P=12$.
I will use drawing values.

Bearing Information

Bearing #	Type	ID (in.)	OD (in.)	Width (in.)	Static Rated Load (lb)	Dynamic Rated Load (lb)	McMaster ID #
1	Ball	1	2	3/8	1,450	2,400	60355K509
2	Ball	1 1/8	2 1/8	3/8		3,850	2780T27
3	Ball	3/4	2	9/16	1,350	2,950	2780T25
4	Ball	3/4	2	9/16	1,350	2,950	2780T25
5	Ball	1	2	3/8	1,450	2,400	60355K509

Shaft Information



Generally better to keep it to only 2 different bearings.
Fewer different parts.

Figure 2. Shaft Graphics

Introduction

The goal of this project is to design a reverted geartrain, the related shafts, and a set of bearings to support the geartrain for a clock tower, converting a single input of 1/60 rpm to outputs of both 1/60 and 1/720 rpm about the same axis. This project will use 1050 Steel quenched and tempered at 800°F as the primary material of all gears, shafts, and spacers, as well as bearings selected from the McMaster-Carr Catalog. Everything must have a functionally infinite life and not come under excessive stress or deflection. Attached at the end of both concentric shafts are two hands, an hour and minute hand dependent on the intended RPM, which apply a torque to their respective shafts.

Solution

Free Body Diagrams:

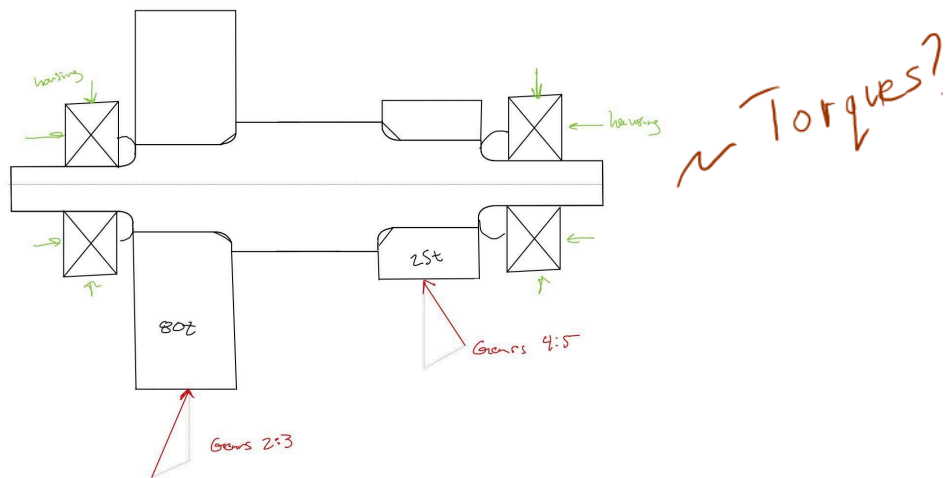


Figure 3. Intermediate Shaft FBD

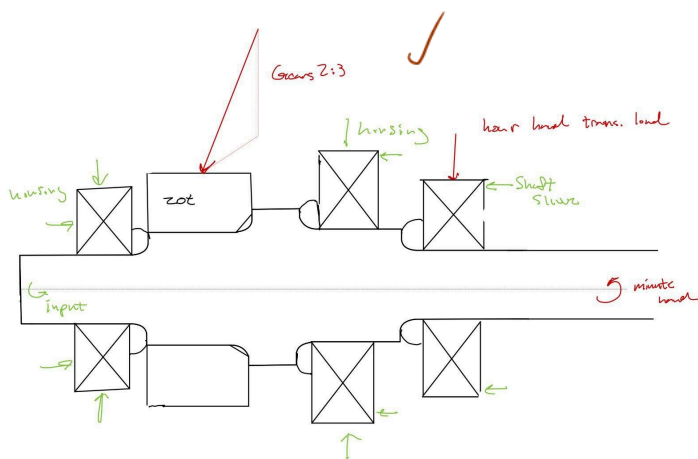


Figure 4. Primary Shaft, Inner FBD

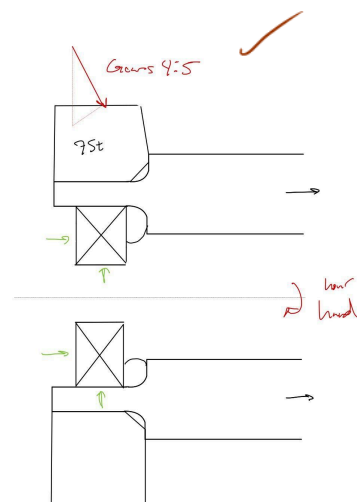
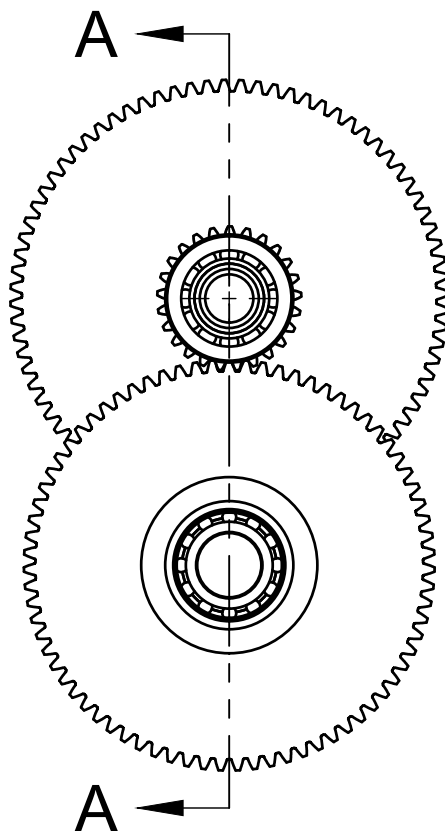
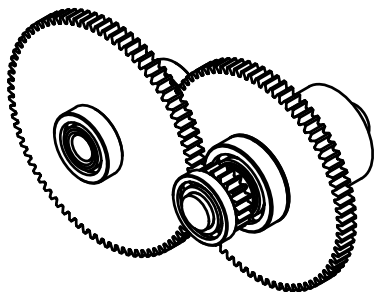


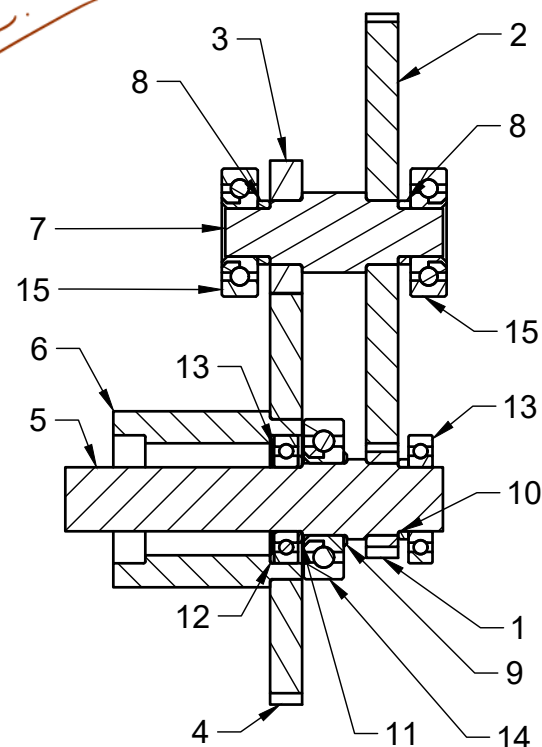
Figure 5. Primary Shaft, Sleeve FBD

Technical Drawings

Parts List	
Title	Number
20t Gear	1
80t Gear	2
25t Gear	3
75t Gear	4
Primary (P.) Shaft, Inner	5
P. Shaft, Sleeve	6
Intermediate (Inter.) Shaft	7
Inter. Shaft, Spacer	8
P. Shaft (P.S.), Inner CS (Clock Side) Spacer	9
P. Shaft (P.S.), Inner IS (Input Side) Spacer	10
P. Shaft (P.S.), Sleeve OF (Outer Face) Spacer	11
P. Shaft (P.S.), Sleeve IF (Inner Face) Spacer	12
60355K509 Ball Bearing (B.B) ✓	13
2780T27 B.B. ✓	14
2780T25 B.B. ✓	15



Nice

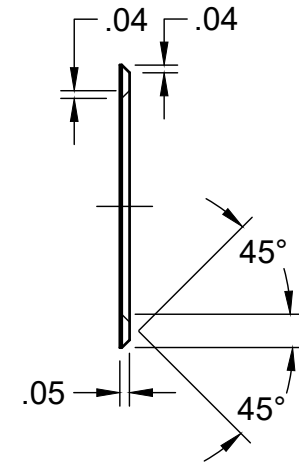
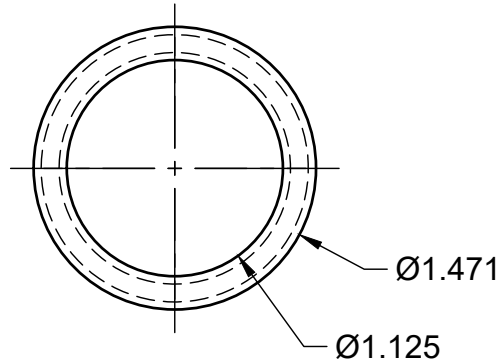
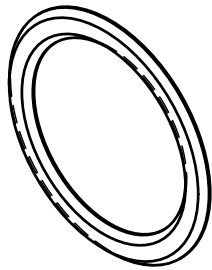


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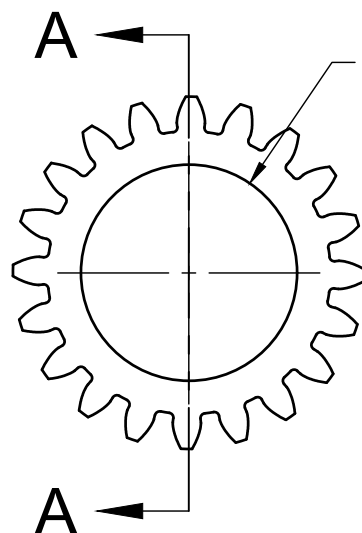
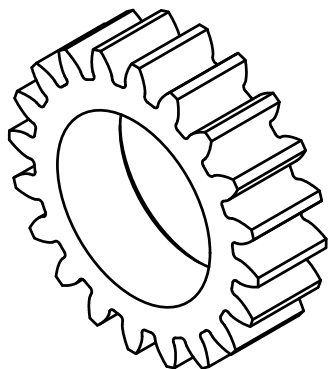
Parts List

Name	Number
20t Gear	1
80t Gear	2
25t Gear	3
75t Gear	4
P. Shaft, Inner	5
P. Shaft, Sleeve	6
Inter. Shaft	7
Inter. Shaft, Spacer	8
P. S. Inner CS Spacer	9
P.S. Inner IS Spacer	10
P.S. Sleeve OF Spacer	11
P.S. Sleeve IF Spacer	12
60355K509 B.B.	13
2780T27 B.B.	14
2780T25 B.B.	15

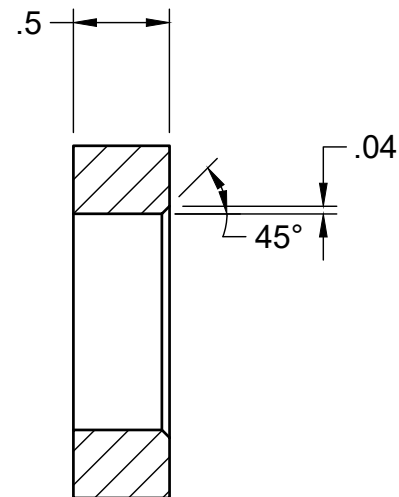
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		Clock Tower Geartrain			
		TITLE			
		General Assembly			
APPROVED	SIZE	CODE	DWG NO		REV
CHECKED	A				
DRAWN	Patrick Jacobs	4/1/2026	SCALE 1:3	WEIGHT	SHEET 1/1



Material	1050 Q&T @ 800F	Unless Otherwise Stated: All Tolerances: 0.005"	PROJECT					
			Clock Tower Geartrain					
			TITLE					
			Primary Shaft, Inner CS Spacer					
APPROVED		SIZE	CODE	DWG NO	REV			
CHECKED								
DRAWN		Patrick Jacobs	4/1/2026	SCALE	1:1	WEIGHT	SHEET	1/1

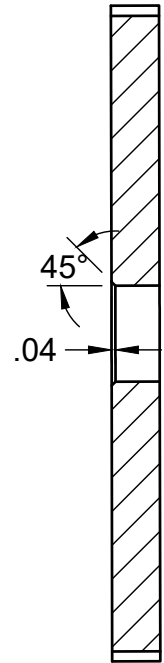
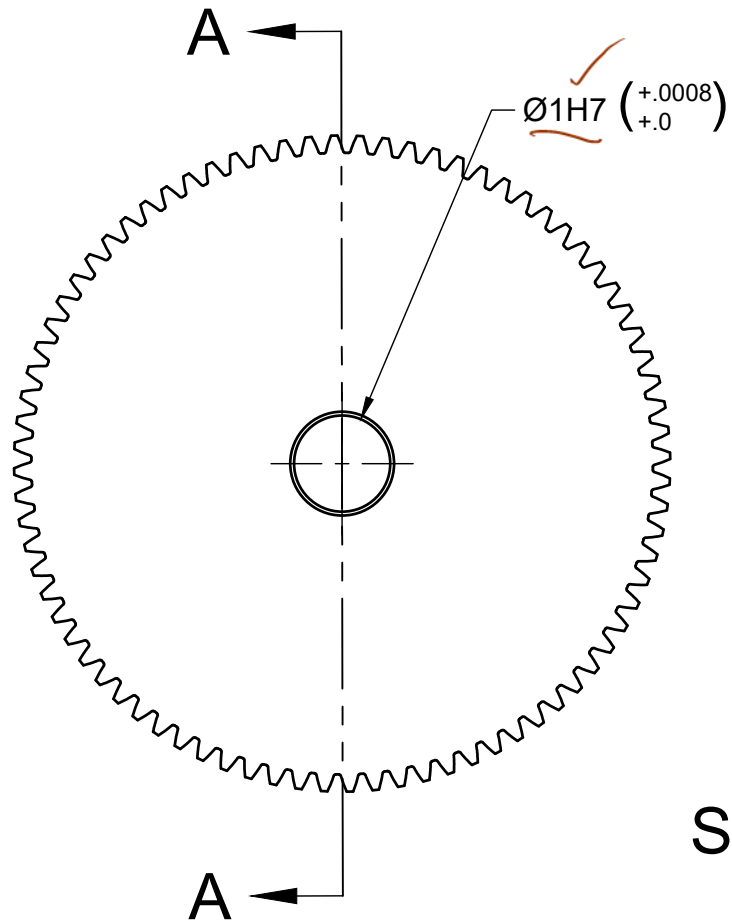
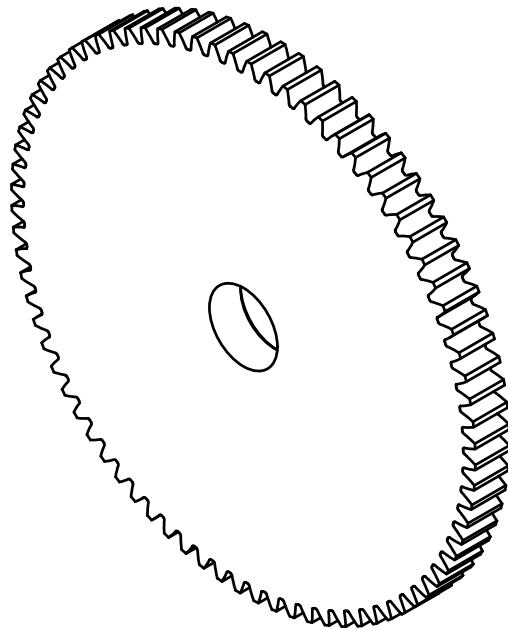


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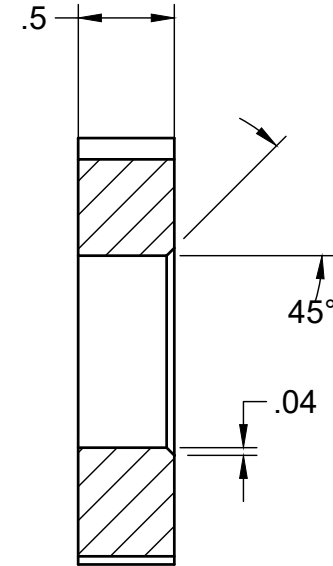
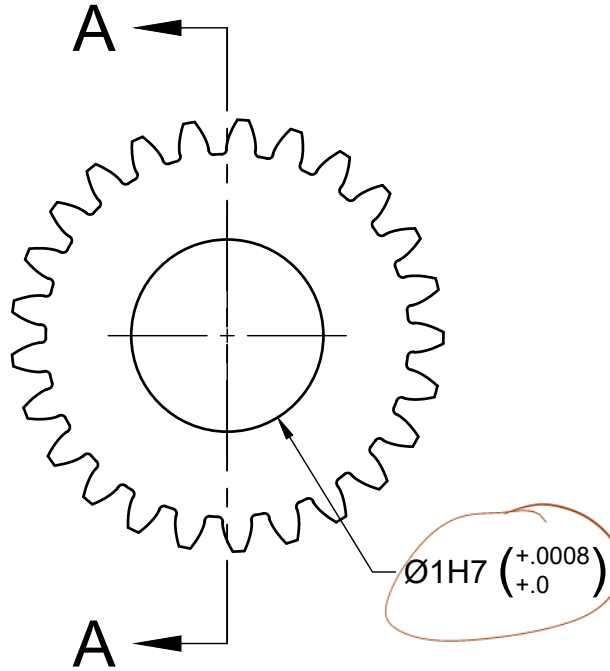
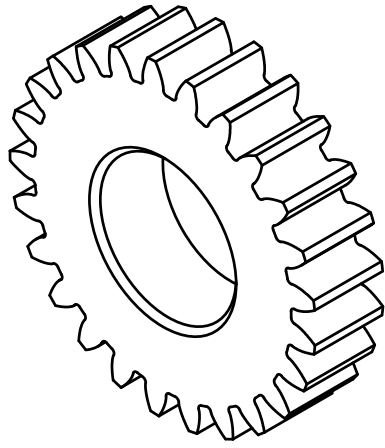
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Teeth	20	Unless Otherwise Stated: All Tolerances: 0.005" ✓	PROJECT				
Pitch	12 tpi		Clock Tower Geartrain				
Face Width	.5 inches		TITLE				
Pressure Angle	20°		CTG - 20t Gear				
Material	1050 Q&T @ 800F						
			APPROVED	SIZE	CODE	DWG NO	REV
			CHECKED	A			
			DRAWN	Patrick Jacobs	4/1/2026	SCALE 1:1 (From parent)	WEIGHT



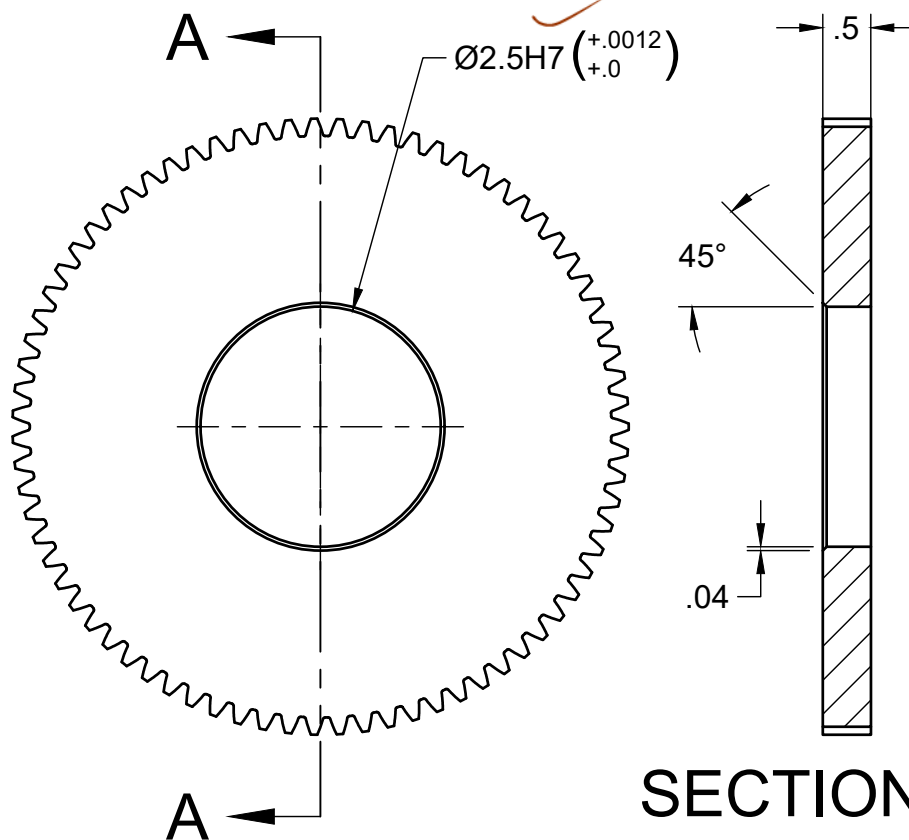
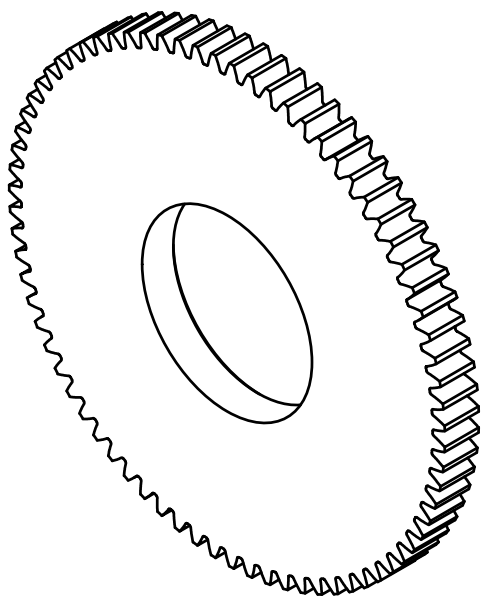
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Teeth	80	Unless Otherwise Stated: All Tolerances: 0.005"	PROJECT			
Pitch	12 tpi		Clock Tower Geartrain			
Face Width	.5 inches		TITLE			
Pressure Angle	20°		CTG - 80t Gear			
Material	1050 Q&T @ 800F					
		APPROVED	SIZE	CODE	DWG NO	REV
		CHECKED	A			
		DRAWN	Patrick Jacobs	4/1/2026	SCALE 1:2	WEIGHT
					SHEET 1/1	



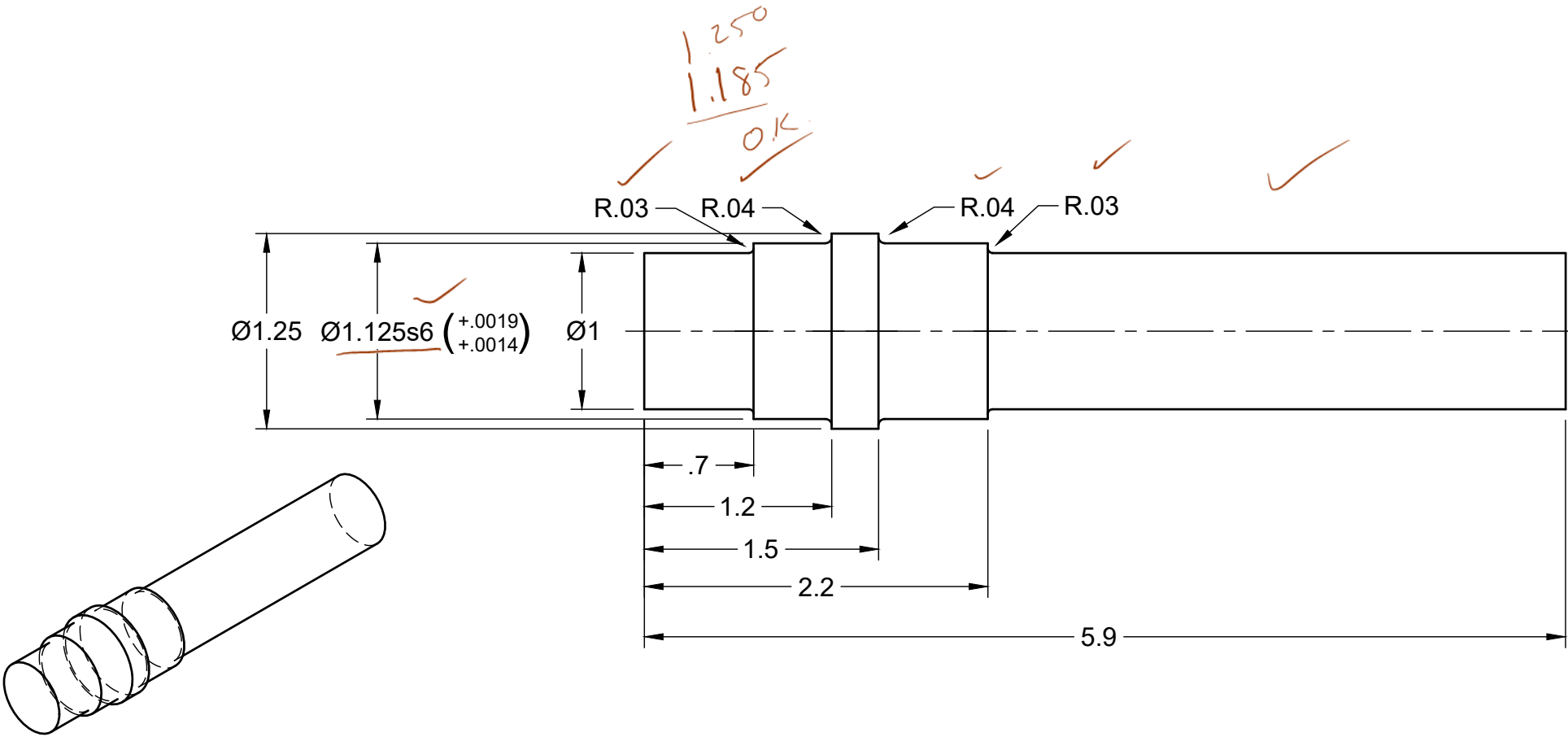
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SCALE 1:1

Teeth	25	Unless Otherwise Stated: All Tolerances: 0.005" ✓	PROJECT			
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Face Width	.5 inches		TITLE			
Pressure Angle	20°		CTG - 25t Gear			
Material	1050 Q&T @ 800F					
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		CHECKED	A			
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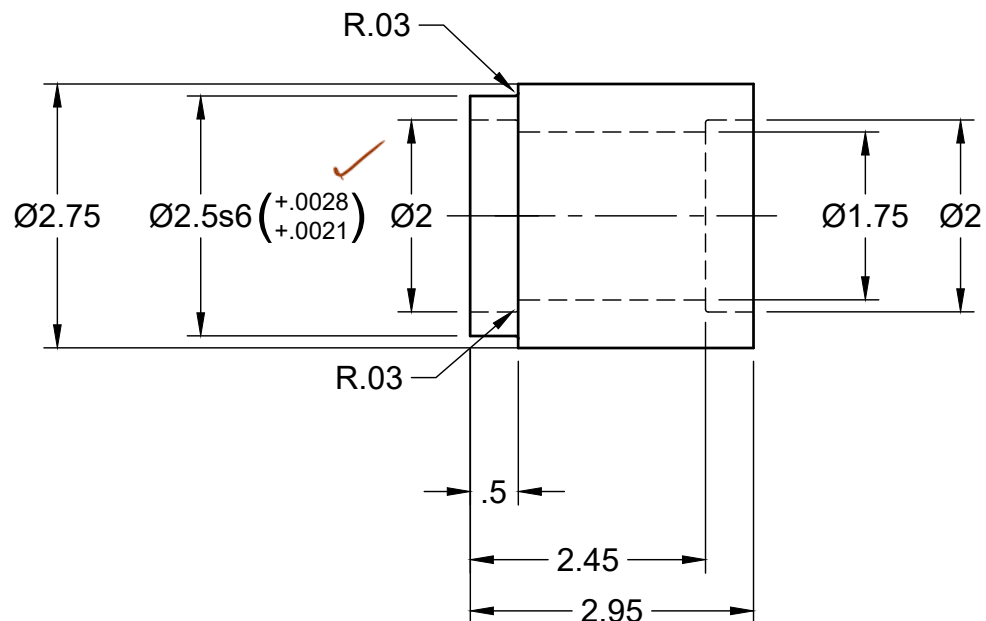
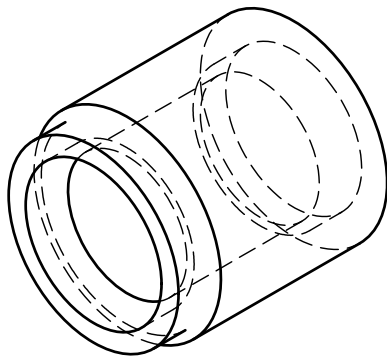


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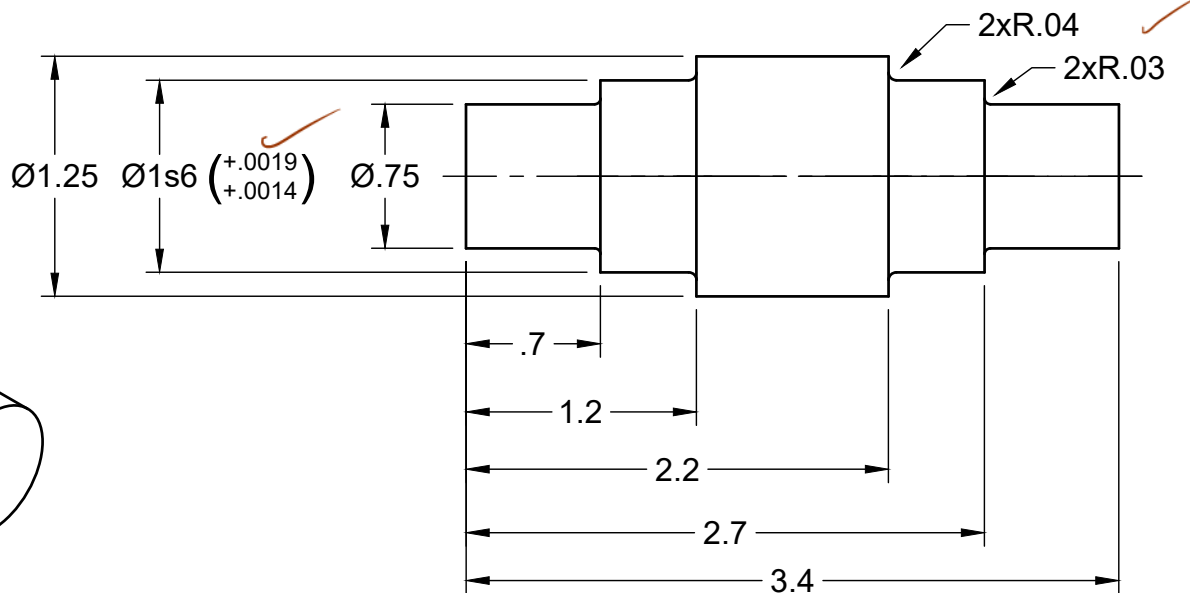
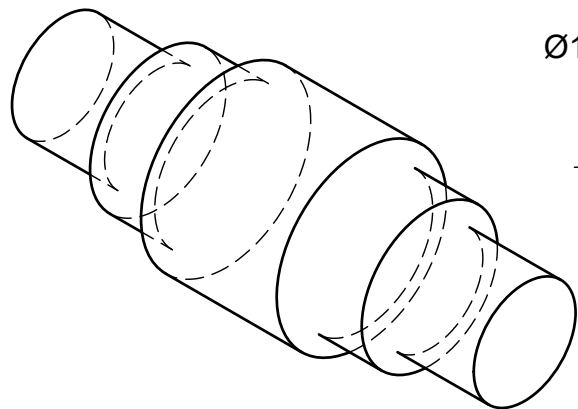
Teeth	75	Unless Otherwise Stated: All Tolerances: 0.005"	PROJECT			
Pitch	12 tpi		Clock Tower Geartrain			
Face Width	.5 inches		TITLE			
Pressure Angle	20°		CTG - 75t Gear			
Material	1050 Q&T @ 800F					
		APPROVED	SIZE	CODE	DWG NO	REV
		CHECKED	A			
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					SHEET 1/1	



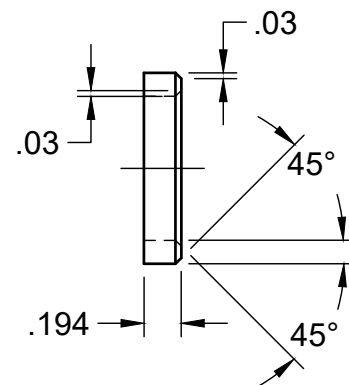
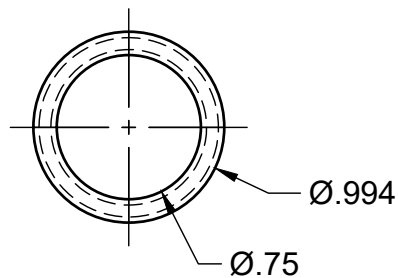
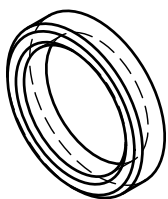
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			TITLE Primary Shaft, Inner			
APPROVED		SIZE A	CODE	DWG NO	REV	
CHECKED						
DRAWN	Patrick Jacobs	4/1/2026	SCALE 1:2	WEIGHT	SHEET 1/1	



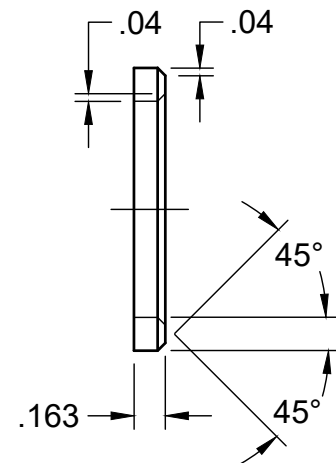
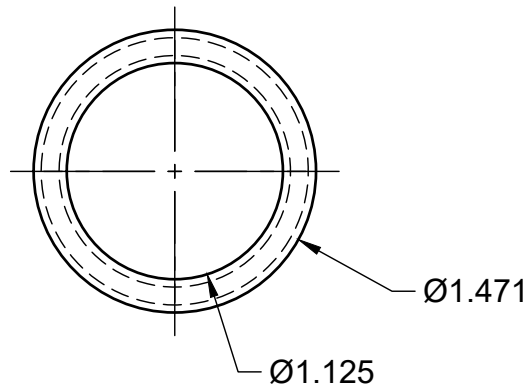
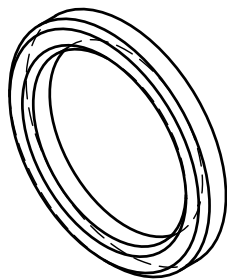
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			TITLE Primary Shaft, Sleeve				
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		CHECKED					
DRAWN Patrick Jacobs 4/1/2026			SCALE 1:2	WEIGHT		SHEET 1/1	



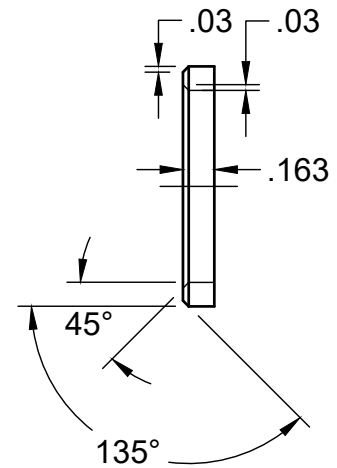
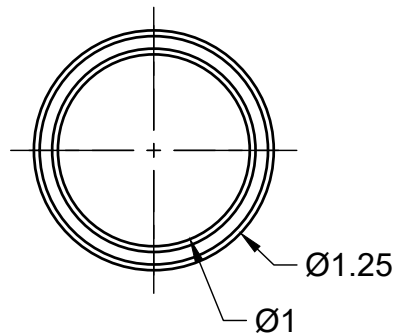
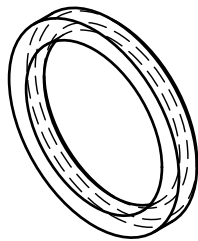
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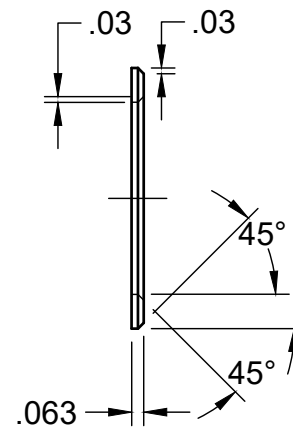
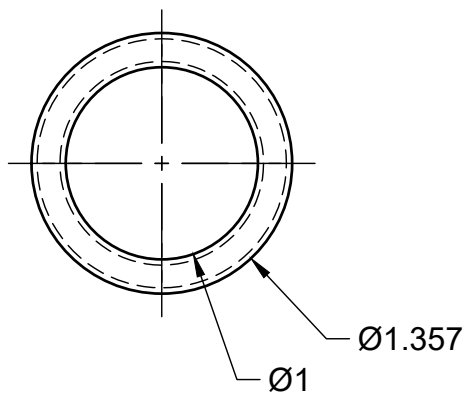
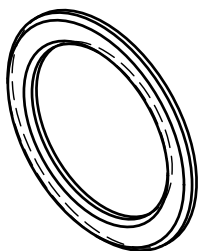
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DRAWN		Patrick Jacobs	4/1/2026	SCALE 1:1	WEIGHT	SHEET 1/1



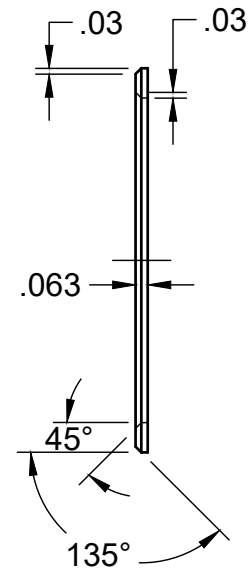
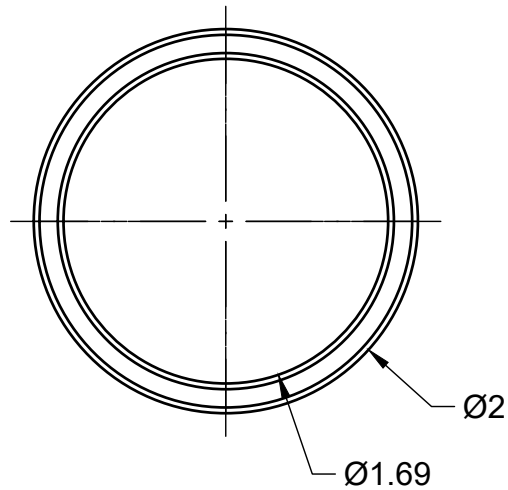
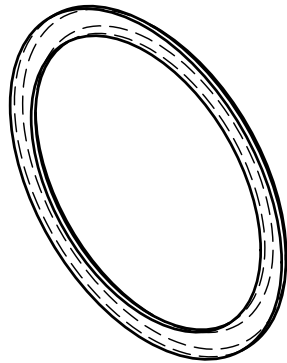
Material	1050 Q&T @ 800F	Unless Otherwise Stated: All Tolerances: 0.005"	PROJECT Clock Tower Geartrain					
			TITLE Primary Shaft, Inner CS Spacer					
			APPROVED	SIZE A	CODE	DWG NO	REV	
			CHECKED					
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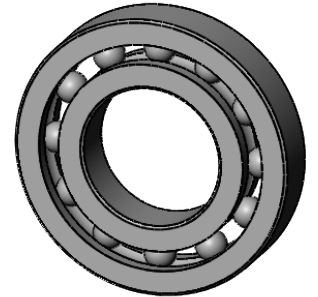
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CHECKED		A				
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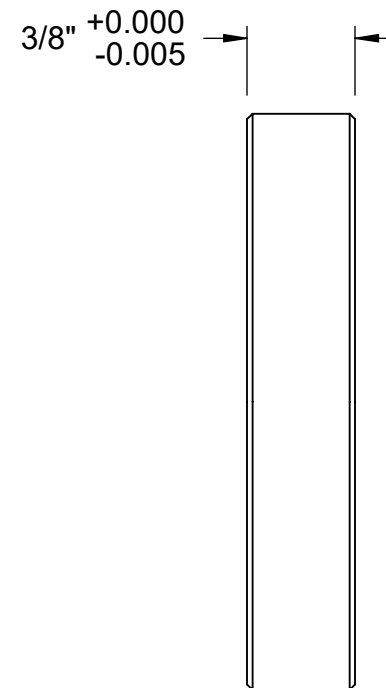
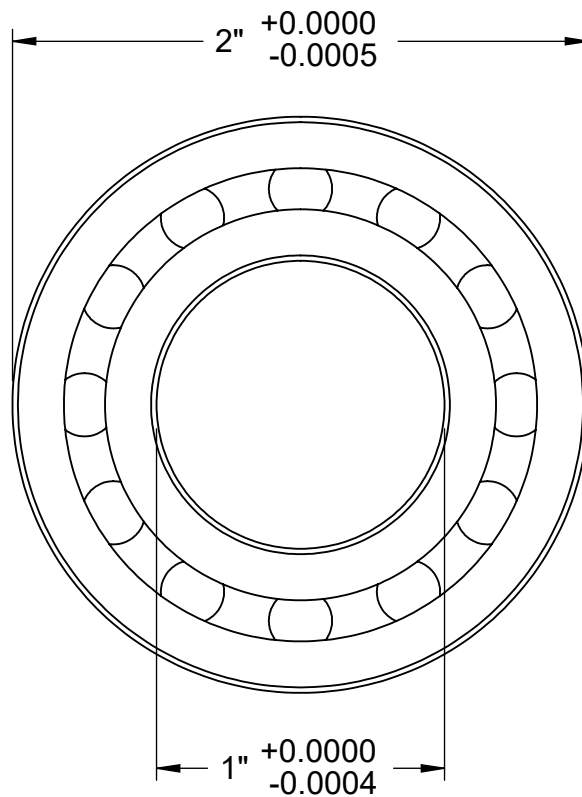
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			TITLE Primary Shaft, Sleeve OF Spacer			
APPROVED		SIZE A	CODE	DWG NO	REV	
CHECKED						
DRAWN	Patrick Jacobs	4/1/2026	SCALE 1:1	WEIGHT	SHEET 1/1	



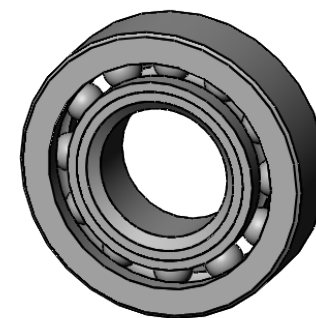
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			TITLE Primary Shaft, Sleeve IF Spacer			
APPROVED			SIZE	CODE	DWG NO	REV
CHECKED		A				
DRAWN Patrick Jacobs 4/1/2026			SCALE 1:1	WEIGHT	SHEET 1/1	



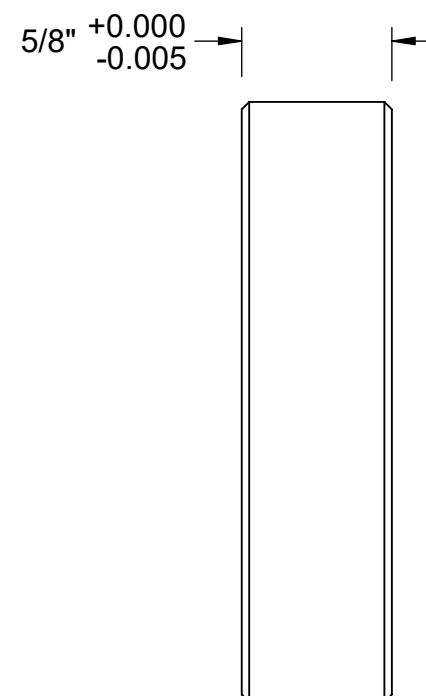
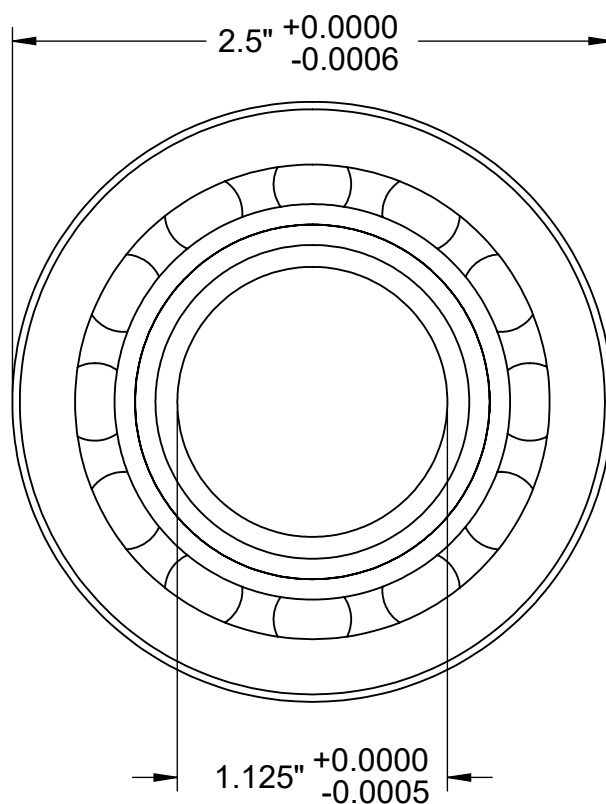
Trade Number: R16



McMASTER-CARR  http://www.mcmaster.com © 2021 McMaster-Carr Supply Company Information in this drawing is provided for reference only.	PART NUMBER	60355K509
	Ball Bearing	



Trade Number: 1652



McMASTER-CARR 

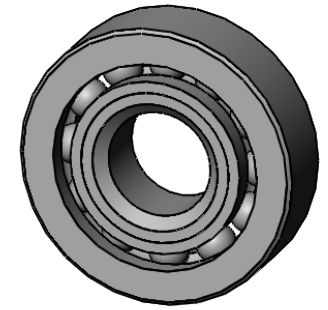
<http://www.mcmaster.com>
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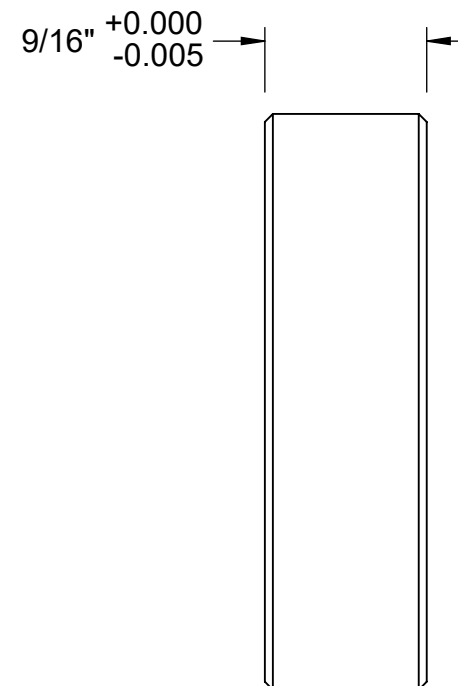
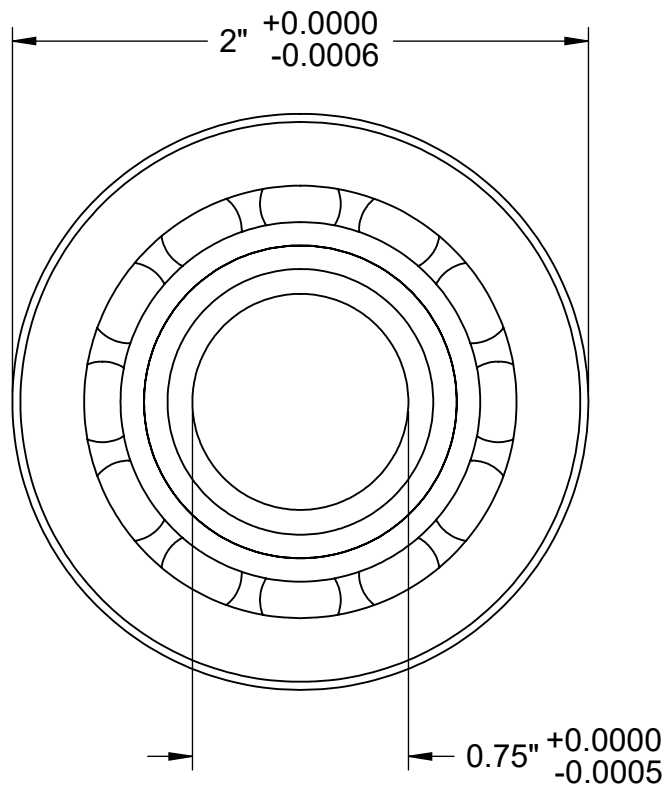
PART
NUMBER

2780T27

High-Load
Ball Bearing



Trade Number: 1638



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Information in this drawing is provided for reference only.

PART
NUMBER

2780T25

High-Load
Ball Bearing

Supporting Materials

Shaft Calculations

Shaft General Drawings

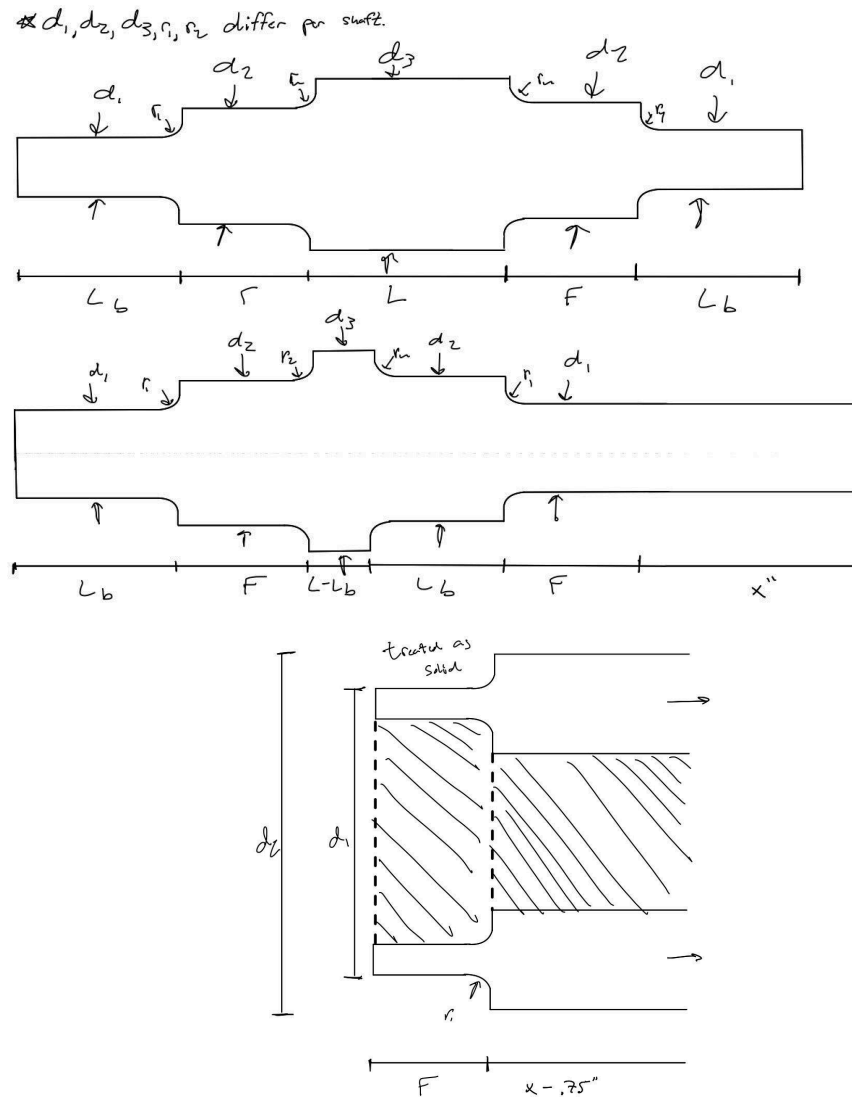


Figure 6. Shaft Drawings with Variable Lengths

Methods of Failure for Shafts

The shafts will fail either via yielding under an initial load or by fatiguing over time. The rotation of the shaft as it is driven will apply an alternating force that produces fatigue, and the weight of the components assembled on each shaft, including the hands at the end, will produce a static load that could result in the shafts yielding under the pressure.

Primary Shaft Calculations

Part I Setup

1 Material Properties 1050 Steel Q&T @800

$$S_y = 115 \text{ ksi}$$

$$S_{ut} = 158 \text{ ksi}$$

2 Dimensions

2.1 Diameters

$$d_1 = 1 \text{ in}$$

$$d_2 = 1.125 \text{ in}$$

$$d_3 = 1.25 \text{ in}$$

2.2 Radii

$$r_1 = 0.03 \text{ in}$$

$$r_2 = 0.04 \text{ in}$$

2.3 Lengths

$$F = 0.5 \text{ in}$$

$$L = 0.3 \text{ in}$$

$$L_B = 0.7 \text{ in}$$

3 Forces

3.1 Gear Forces

Note: W_t from Gear Calculations

$$W_{t2} = 43.446 \text{ lb} \sim 42.2 \checkmark$$
$$W_2 = \frac{W_{t2}}{\cos(20)} = 106.463 \text{ lb} \leftarrow 44.9 \# \text{ X}$$
$$W_{t5} = 139.026 \text{ lb} \sim 135 \# \checkmark$$
$$W_5 = \frac{W_{t5}}{\cos(20)} = 340.683 \text{ lb} \sim 143.7 \# \text{ X}$$

← How?
Use deg, not
radians.

3.2 Bearing Reaction Forces

$$B_2 = \frac{1}{L_B + F + L} \left(W_2 \left(\frac{L_B}{2} + \frac{F}{2} \right) + W_5 \left(\frac{3L_B}{2} + \frac{3F}{2} + L \right) \right) = 519.541 \text{ lb}$$
$$B_1 = W_2 + W_5 - B_2 = -72.395 \text{ lb}$$

4 Torque

Note: T_{min} and T_2 from Gear Calculations

$$T_{min} = 1426.01 \text{ in} \cdot \text{lb} \quad 1425 \checkmark$$
$$T_2 = 36.205 \text{ in} \cdot \text{lb}$$
$$T = T_{min} + T_2 = 1462.22 \text{ in} \cdot \text{lb} \quad 1461 \checkmark$$

Part II Calculations

5 Shear and Moment Diagram

5.1 Intermediate Calculations

$$I_{nt} = B_1 - W_2 = -178.858 \text{ lb}$$
$$A_1 = B_1 \left(\frac{L_B}{2} + \frac{F}{2} \right) = -43.437 \text{ in} \cdot \text{lb}$$
$$A_2 = I_{nt} \left(\frac{F}{2} + L + \frac{L_B}{2} \right) + A_1 = -204.410 \text{ in} \cdot \text{lb}$$

5.2 Moments at Stress Concentrations $\checkmark$

$$M_1 = A_2 + W_5 \left(\frac{L_B}{2} \right) = -85.171 \text{ in} \cdot \text{lb}$$
$$M_2 = A_1 + I_{nt} \left(\frac{F}{2} + L \right) = -141.809 \text{ in} \cdot \text{lb}$$

6 Stress Risers at First Step

6.1 Ratios

$$\frac{r_1}{d_1} = 0.03$$

$$\frac{d_2}{d_1} = 1.125$$

Bearing Force from gear.

?
A diagram would be helpful.

What are you doing?

} ?

6.2 Bending Stress Riser

Note: Figure A-15-9

$$K_{t1} = 2.1$$

$$a_b = 0.246 - 3.08 \cdot 10^{-3} S_{ut} + 1.51 \cdot 10^{-5} S_{ut}^2 - 2.67 \cdot 10^{-8} S_{ut}^3 = 0.03100$$

$$\underline{q_{b1}} = \frac{1}{1 + \frac{a_b}{\sqrt{r_1}}} = 0.8482$$

$$K_{f1} = 1 + q_{b1} (K_{t1} - 1) = \underline{1.933} \text{ ,ok.}$$

6.3 Torsion Stress Riser

Note: Figure A-15-8

$$K_{ts1} = 1.6$$

$$a_t = 0.19 - 2.51 \cdot 10^{-3} S_{ut} + 1.35 \cdot 10^{-5} S_{ut}^2 - 2.67 \cdot 10^{-8} S_{ut}^3 = 0.02512$$

$$q_{t1} = \frac{1}{1 + \frac{a_t}{\sqrt{r_1}}} = 0.8733$$

$$K_{fs1} = 1 + q_{t1} (K_{ts1} - 1) = 1.524 \quad \checkmark$$

7 Stress Riser at Second Step

7.1 Ratios

$$\frac{r_2}{d_2} = 0.03556$$

$$\frac{d_3}{d_2} = 1.111$$

7.2 Bending Stress Riser

Note: Figure A-15-8

$$K_{t2} = 2$$

$$q_{b2} = \frac{1}{1 + \frac{a_b}{\sqrt{r_2}}} = 0.8658$$

$$K_{f2} = 1 + q_{b2} (K_{t2} - 1) = \underline{1.866} \quad \checkmark$$

7.3 Torsion Stress Riser

Note: Figure A-15-8

$$K_{ts2} = 1.6$$

$$q_{t2} = \frac{1}{1 + \frac{a_t}{\sqrt{r_2}}} = 0.8884$$

$$K_{fs2} = 1 + q_{t2} (K_{ts2} - 1) = 1.533 \quad \checkmark$$

8 Stresses at Concentrations

8.1 First Step

$$\sigma_{a1} = \left(\left(K_{f1} \frac{32M_1}{\pi d_1^3} \right)^2 + 3 \left(K_{fs1} \frac{16T}{\pi d_1^3} \right)^2 \right)^{\frac{1}{2}} = 19729 \text{ psi}$$

8.2 Second Step

$$\sigma_{a2} = \left(\left(K_{f2} \frac{32M_2}{\pi d_2^3} \right)^2 + 3 \left(K_{fs2} \frac{16T}{\pi d_2^3} \right)^2 \right)^{\frac{1}{2}} = 14016 \text{ psi}$$

9 Endurance Strength

9.1 Surface Factor (Machined)

$$a = 2$$

$$b = -0.217$$

$$k_a = aS_{ut}^b = 0.6667$$

9.2 Size Factor

$$k_{b1} = \left(\frac{d_1}{0.3} \right)^{-0.107} = 0.8791$$

$$k_{b2} = \left(\frac{d_2}{0.3} \right)^{-0.107} = 0.8681$$

9.3 Temperature Factor (Ambient)

$$k_d = 1$$

9.4 Reliability Factor (99%, Table 6-4)

$$k_e = 0.814$$

9.5 Endurance Strength

9.5.1 First Step

$$S_{e1} = k_a k_{b1} k_d k_e \cdot \frac{S_{ut}}{2} = 37.690 \text{ ksi}$$

9.5.2 Second Step

$$S_{e2} = k_a k_{b2} k_d k_e \cdot \frac{S_{ut}}{2} = 37.218 \text{ ksi}$$

10 Safety Factor

10.1 First Step

10.1.1 Goodman

$$n_{f1} = \frac{S_{e1} \cdot 10^3}{\sigma_{a1}} = 1.910$$

Good.

10.1.2 Langer Lines

$$n_{y1} = \frac{S_y \cdot 10^3}{\sigma_{a1}} = 5.829$$

10.2 Second Step

10.2.1 Goodman

$$n_{f2} = \frac{S_{e2} \cdot 10^3}{\sigma_{a2}} = 2.655$$

10.2.2 Langer Lines

$$n_{y2} = \frac{S_y \cdot 10^3}{\sigma_{a2}} = 8.205$$

Intermediate Shaft Calculations

Part I Setup

1 Material Properties 1050 Steel Q&T @800

$$S_y = 115 \text{ ksi}$$

$$S_{ut} = 158 \text{ ksi}$$

2 Dimensions

2.1 Diameters

$$d_1 = 0.75 \text{ in}$$

$$d_2 = 1 \text{ in}$$

$$d_3 = 1.25 \text{ in}$$

2.2 Radii

$$r_1 = 0.03 \text{ in}$$

$$r_2 = 0.04 \text{ in}$$

2.3 Lengths

$$F = 0.5 \text{ in}$$

$$L = 1 \text{ in}$$

$$L_B = 0.7 \text{ in}$$

3 Forces

3.1 Gear Forces

Note: W_t from Gear Calculations

$$W_{t4} = 139.026 \text{ lb}$$

$$W_{t3} = 43.446 \text{ lb}$$

$$W_4 = \frac{W_{t4}}{\cos(20)} = 340.683 \text{ lb}$$

$$W_3 = \frac{W_{t3}}{\cos(20)} = 106.463 \text{ lb}$$

3.2 Bearing Reaction Forces

$$B_4 = \frac{1}{L_B + 2F + L} \left(W_3 \left(\frac{L_B}{2} + \frac{F}{2} \right) + W_4 \left(\frac{L_B}{2} + \frac{3F}{2} + L \right) \right) = 288.634 \text{ lb}$$
$$B_3 = W_3 + W_4 - B_4 = 158.512 \text{ lb}$$

4 Torque

Note: $T = T_3$ from Gear Calculations

$$T = 144.819 \text{ in} \cdot \text{lb}$$

140 in.lbs. O.K.

Part II

Calculations

5 Shear and Moment Diagram

5.1 Intermediate Calculations

$$I_{nt} = B_3 - W_3 = 52.049 \text{ lb}$$
$$A_1 = B_3 \left(\frac{L_B}{2} + \frac{F}{2} \right) = 95.107 \text{ in} \cdot \text{lb}$$
$$A_2 = I_{nt} (F + L) + A_1 = 173.180 \text{ in} \cdot \text{lb}$$

5.2 Moments at Stress Concentrations

$$M_1 = A_2 - B_4 \left(\frac{F}{2} \right) = 101.022 \text{ in} \cdot \text{lb}$$
$$M_2 = A_1 + I_{nt} \left(\frac{F}{2} + L \right) = 160.168 \text{ in} \cdot \text{lb}$$

6 Stress Risers at First Step

6.1 Ratios

$$\frac{r_1}{d_1} = 0.04$$

$$\frac{d_2}{d_1} = 1.333$$

6.2 Bending Stress Riser

Note: Figure A-15-9

$$K_{t1} = 1.8$$
$$a_b = 0.246 - 3.08 \cdot 10^{-3} S_{ut} + 1.51 \cdot 10^{-5} S_{ut}^2 - 2.67 \cdot 10^{-8} S_{ut}^3 = 0.03100$$
$$q_{b1} = \frac{1}{1 + \frac{a_b}{\sqrt{r_1}}} = 0.8482$$
$$K_{f1} = 1 + q_{b1} (K_{t1} - 1) = 1.679$$

6.3 Torsion Stress Riser

Note: Figure A-15-8

$$\begin{aligned}K_{ts1} &= 2.1 \\a_t &= 0.19 - 2.51 \cdot 10^{-3} S_{ut} + 1.35 \cdot 10^{-5} S_{ut}^2 - 2.67 \cdot 10^{-8} S_{ut}^3 = 0.02512 \\q_{t1} &= \frac{1}{1 + \frac{a_t}{\sqrt{r_1}}} = 0.8733 \\K_{fs1} &= 1 + q_{t1} (K_{ts1} - 1) = 1.961\end{aligned}$$

7 Stress Riser at Second Step

7.1 Ratios

$$\begin{aligned}\frac{r_2}{d_2} &= 0.04 \\\frac{d_3}{d_2} &= 1.25\end{aligned}$$

7.2 Bending Stress Riser

Note: Figure A-15-8

$$\begin{aligned}K_{t2} &= 1.7 \\q_{b2} &= \frac{1}{1 + \frac{a_b}{\sqrt{r_2}}} = 0.8658 \\K_{fs2} &= 1 + q_{b2} (K_{t2} - 1) = 1.606\end{aligned}$$

7.3 Torsion Stress Riser

Note: Figure A-15-8

$$\begin{aligned}K_{ts2} &= 1.5 \\q_{t2} &= \frac{1}{1 + \frac{a_t}{\sqrt{r_2}}} = 0.8884 \\K_{fs2} &= 1 + q_{t2} (K_{ts2} - 1) = 1.444\end{aligned}$$

8 Stresses at Concentrations

8.1 First Step

$$\sigma_{a1} = \left(\left(K_{f1} \frac{32M_1}{\pi d_1^3} \right)^2 + 3 \left(K_{fs1} \frac{16T}{\pi d_1^3} \right)^2 \right)^{\frac{1}{2}} = 7212 \text{ psi}$$

8.2 Second Step

$$\sigma_{a2} = \left(\left(K_{f2} \frac{32M_2}{\pi d_2^3} \right)^2 + 3 \left(K_{fs2} \frac{16T}{\pi d_2^3} \right)^2 \right)^{\frac{1}{2}} = 3205 \text{ psi}$$

9 Endurance Strength

9.1 Surface Factor (Machined)

$$a = 2$$

$$b = -0.217$$

$$k_a = aS_{ut}^b = 0.6667$$

9.2 Size Factor

$$k_{b1} = \left(\frac{d_1}{0.33} \right)^{-0.107} = 0.9159$$

$$k_{b2} = \left(\frac{d_2}{0.3} \right)^{-0.107} = 0.8791$$

9.3 Temperature Factor (Ambient)

$$k_d = 1 = 1$$

9.4 Reliability Factor (99%, Table 6-4)

$$k_e = 0.814$$

9.5 Endurance Strength

9.5.1 First Step

$$S_{e1} = k_a k_{b1} k_d k_e \cdot \frac{S_{ut}}{2} = 39.267 \text{ ksi}$$

9.5.2 Second Step

$$S_{e2} = k_a k_{b2} k_d k_e \cdot \frac{S_{ut}}{2} = 37.690 \text{ ksi}$$

10 Safety Factor

10.1 First Step

10.1.1 Goodman

$$n_{f1} = \frac{S_{e1} \cdot 10^3}{\sigma_{a1}} = 5.445$$

10.1.2 Langer Lines

$$n_{y1} = \frac{S_y \cdot 10^3}{\sigma_{a1}} = 15.946$$

10.2 Second Step

10.2.1 Goodman

$$n_{f2} = \frac{S_{e2} \cdot 10^3}{\sigma_{a2}} = 11.761$$

10.2.2 Langer Lines

$$n_{y2} = \frac{S_y \cdot 10^3}{\sigma_{a2}} = 35.886$$

Sleeve Calculations

Part I Setup

1 Material Properties 1050 Steel Q&T @800

$$S_y = 82 \text{ ksi}$$

$$S_{ut} = 108 \text{ ksi}$$

2 Dimensions

2.1 Diameters

$$d_1 = 2.5 \text{ in}$$

$$d_2 = 3.5 \text{ in}$$

These are quite large, partially due to calculation errors.

2.2 Radius

$$r_1 = 0.03 \text{ in} \checkmark$$

3 Torque

Note: $T = T_{hour}$ from Gear Calculations

$$T = 434.458 \text{ in} \cdot \text{lb}$$

422 in lb, OK.

Part II Calculations

4 Stress Risers

4.1 Ratios

$$\frac{r_1}{d_1} = 0.012$$

$$\frac{d_2}{d_1} = 1.4$$

4.2 Torsion Stress Riser

Note: Figure A-15-8

$$\begin{aligned}K_{ts1} &= 2.8 \\a_t &= 0.19 - 2.51 \cdot 10^{-3} S_{ut} + 1.35 \cdot 10^{-5} S_{ut}^2 - 2.67 \cdot 10^{-8} S_{ut}^3 = 0.04275 \\q_{t1} &= \frac{1}{1 + \frac{a_t}{\sqrt{r_1}}} = 0.8020 \\K_{fs1} &= 1 + q_{t1} (K_{ts1} - 1) = 2.444\end{aligned}$$

5 Stresses at Concentration

$$\sigma_{a1} = \left(+3 \left(K_{fs1} \frac{16T}{\pi d_1^3} \right)^2 \right)^{\frac{1}{2}} = 599.4 \text{ psi}$$

6 Endurance Strength

6.1 Surface Factor (Machined)

$$\begin{aligned}a &= 2 \\b &= -0.217 \\k_a &= a S_{ut}^b = 0.7241\end{aligned}$$

6.2 Size Factor

$$k_b = 0.91 d_1^{-0.157} = 0.7881$$

6.3 Temperature Factor (Ambient)

$$k_d = 1 = 1$$

6.4 Reliability Factor (99%, Table 6-4)

$$k_e = 0.814 = 0.814$$

6.5 Endurance Strength

$$S_{e1} = k_a k_b k_d k_e \cdot \frac{S_{ut}}{2} = 25.082 \text{ ksi}$$

7 Safety Factor

7.1 Goodman

$$n_{f1} = \frac{S_{e1} \cdot 10^3}{\sigma_{a1}} = 41.846$$

7.1.1 Langer Lines

$$n_{y1} = \frac{S_y \cdot 10^3}{\sigma_{a1}} = 136.81$$

Bearings Calculations

Methods of Failure for Bearings

The three main modes of failure for the bearings include static loading, dynamic loading, and exceeding the maximum angular misalignment for the respective bearing. All three of these were accounted for in the calculations below. Note, dynamic load was found assuming a factor of safety of 1.5. It was also assumed that the bearings can still function normally up until a shaft reaches an angle of deflection of 0.003 radians. Additionally, the catalog number and McMaster-Carr page of each bearing used can be found in the executive summary section above.

Static Loading on Bearings



<i>Bearing #</i>	<i>Static Load (lb)</i>	<i>Static Rated Load (lb)</i>	<i>Will This Hold Up?</i>
1	70.395	1,450	Yes
2	519.541	2,100	Yes
3	158.512	1,350	Yes
4	288.634	1,350	Yes
5	340.683	1,450	Yes

Dynamic Loading on Bearings



<i>Bearing #</i>	<i>Dynamic Load (lb)</i>	<i>Dynamic Rated Load (lb)</i>	<i>Will his Hold Up?</i>
1	456.080	2,400	Yes
2	3366.037	3,850	Yes
3	1026.978	2,950	Yes
4	1870.022	2,950	Yes
5	2207.240	2,400	Yes

Max allowable dynamic load:

$$L_d = 8.736 \times 10^6 \text{ cycles}$$

$L_r = 10^6$ cycles $\rightarrow$ ball bearings

$$X_0 = \frac{L_d}{L_r} = 8.736$$

$$\alpha_f = 1.50 \rightarrow \text{FOS}$$

$$X_0 = 0.02$$

$$\Theta = 4.459$$

$$R_0 = 0.997 \rightarrow \text{reliability}$$

$$a = 3, b = 1.483$$

$$C_{10} = \alpha_f \cdot F_0 \left(\frac{X_0}{X_0 + (\Theta - X_0) \left(\ln \left(\frac{1}{R_0} \right)^{1/b} \right)} \right)^{1/a}$$

When $F_0 = 519.541$ ^{lb} $\rightarrow$ worst bearing case bearing 2

$$C_{10} = 3366.037 \text{ lb}$$

C_{10} is less than max allowable for bearing 2, will not fail.

★ Note repeated this math for every bearing seen in the table below. ↓↓

Figure 7. Dynamic Load Calculations

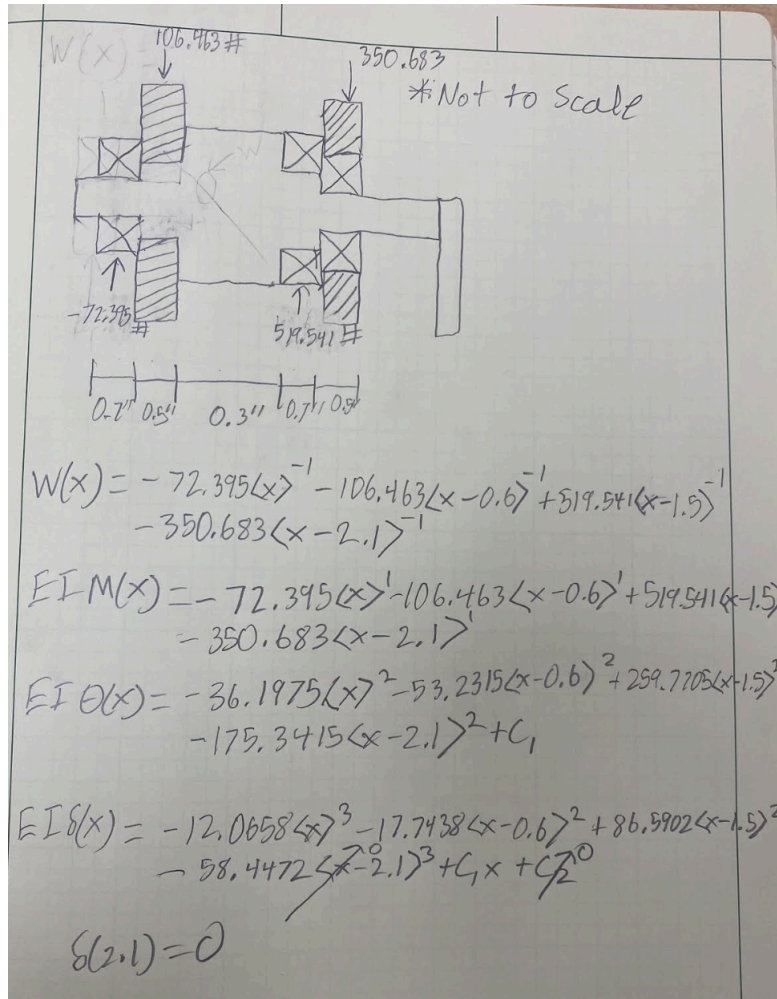


Figure 8. Primary Shaft Slope Calculations

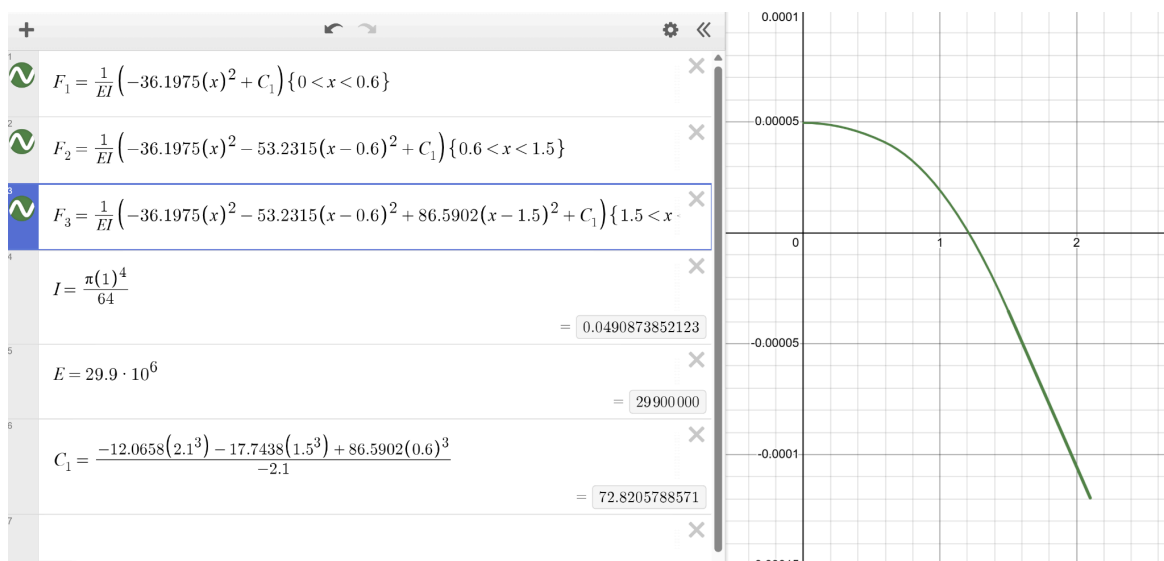


Figure 9. Primary Shaft Slope Calculations. Maximum ~ 0.00011 rad at right end.

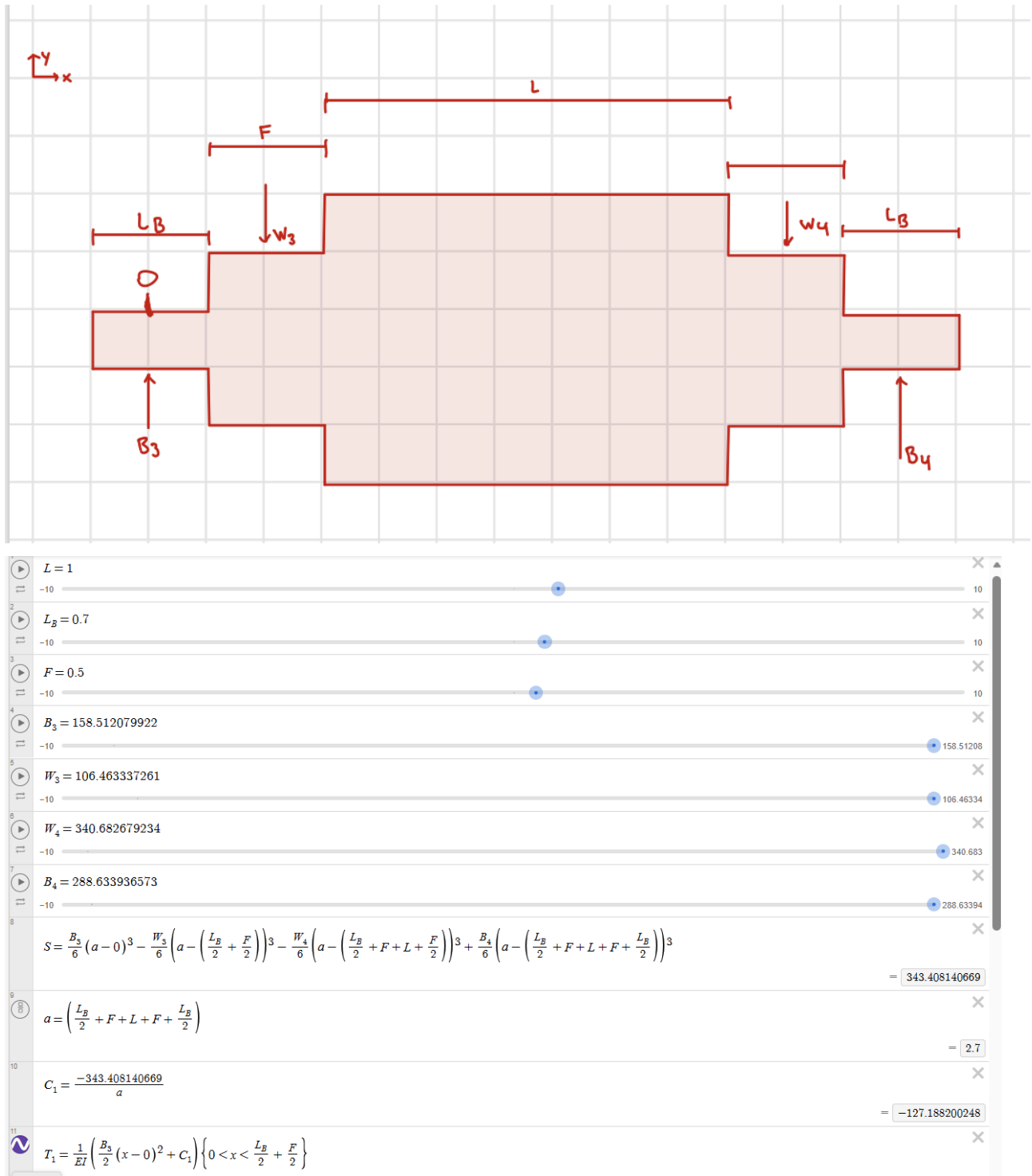


Figure 10. Intermediate Shaft Slope Calculations

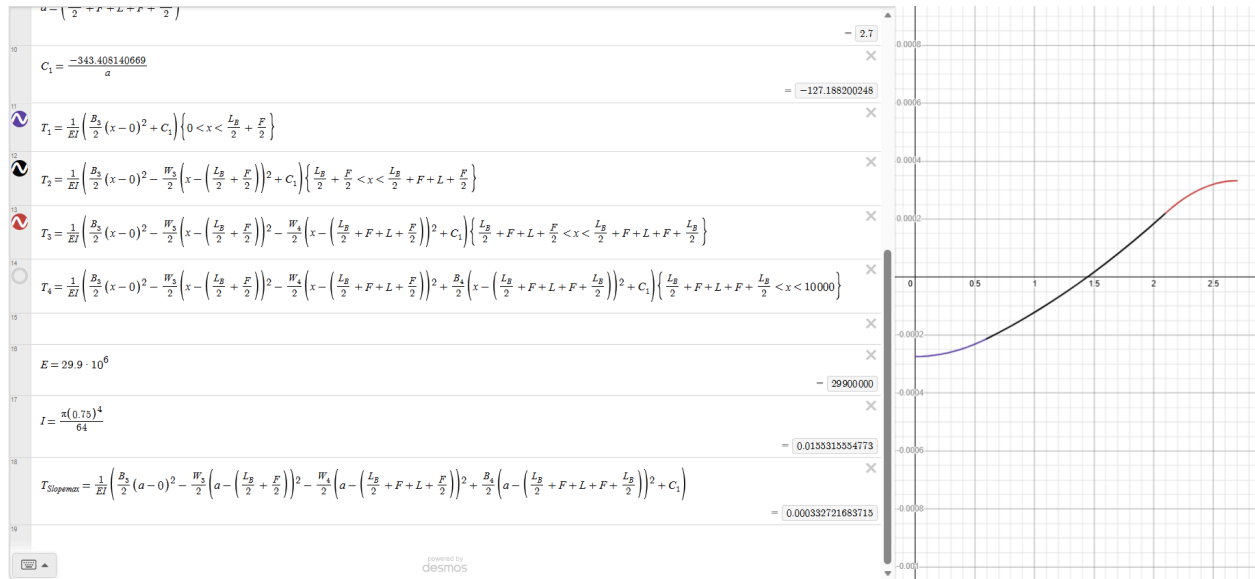


Figure 11. Intermediate Shaft Slope Calculations, Maximum ~ 0.000333 rad at right end.

Gear Calculations

Methods of Failure for Gears

The primary modes of failure for the gear are via fatiguing of the teeth or excessive contact stresses between the teeth. These are derived out of the force that each gear applies on the gear it is mated with. Bending forces on the tooth will cause it to eventually fracture at the tooth's root, tearing the tooth off and rendering the gear inoperable. Contact stress is caused by the compressive contact stresses at the tooth's surface, slowly chipping away at the tooth's surface in what is known as pitting. This results in excessive backlash, gear skippage, and eventually the gear becomes inoperable.

The following calculations include both gear fatigue and contact stress as well as the press fit calculations to determine the diameters of the gear-shaft interfaces. These tolerances were included in the technical drawings above.

Press Fit Checks

1 Material Properties 1050 Steel Q&T @800

$$E = 29.9 \cdot 10^6 \text{ psi}$$

Note: Fairly conservative estimate for friction

$$\mu_s = 0.4$$

2 Check for Primary Shaft Gear (N2)

2.1 Base Dimensions

$$L = 0.5 \text{ in}$$

$$d = 1.125 \text{ in}$$

$$d_o = 1.667 \text{ in}$$

$$d_i = 0 \text{ in}$$

Note: Spec for 1.125H7/s6

2.2 Fit Dimensions

$$\delta_F = 0.0017 \text{ in}$$

$$d_{min} = d + \delta_F = 1.1267 \text{ in}$$

$$\Delta_D = 0.001 \text{ in}$$

$$D_{max} = d + \Delta_D = 1.126 \text{ in}$$

$$\delta_{min} = d_{min} - D_{max} = 0.0007 \text{ in}$$

2.3 Pressure

$$p = \frac{E\delta_{min}}{2d^3} \left(\frac{(d_o^2 - d^2)(d^2 - d_i^2)}{d_o^2 - d_i^2} \right) = 5064 \text{ psi}$$

2.4 Max Torque

$$T = \frac{\pi}{2} \mu_s p L d = 1790 \text{ in} \cdot \text{lb}$$

3 Check for Intermediate Shaft Gears (N3,N4)

3.1 Base Dimensions

$$L = 0.5 \text{ in}$$

$$d = 1 \text{ in}$$

$$d_o = 2.083 \text{ in}$$

$$d_i = 0 \text{ in}$$

Note: 1H7/s6

3.2 Fit Dimensions

$$\delta_F = 0.0014 \text{ in}$$

$$d_{min} = d + \delta_F = 1.0014 \text{ in}$$

$$\Delta_D = 0.0008 \text{ in}$$

$$D_{max} = d + \Delta_D = 1.0008 \text{ in}$$

$$\delta_{min} = d_{min} - D_{max} = 0.0006 \text{ in}$$

.0006" I think the calculation was correct (.0006"), but you put the wrong number (.0014") here.

3.3 Pressure

$$p = \frac{E\delta_{min}}{2d^3} \left(\frac{(d_o^2 - d^2)(d^2 - d_i^2)}{d_o^2 - d_i^2} \right) = 6803 \text{ psi}$$

P = 6694 ok.
P = 15621 with $\delta = .0014$ "

3.4 Max Torque

$$A = \pi d \cdot l$$

$$T = \frac{\pi}{2} \mu_s p L d = 2169 \text{ in} \cdot \text{lb}$$

$$T = F \cdot r = \mu(p \cdot A) \cdot r = \mu p \cdot \pi d \cdot \delta / 2 = \frac{\pi d^2}{2} \mu p$$

T = 20,448 in/lb.
with P = 6694. & $\mu = .8$ (or .5)
Steel on steel, machined
is $\mu = .8$. If polished,
then $\mu = .3$ to .5.
Will still hold.

4 Check for Sleeve Gear (N5)

4.1 Base Dimensions

$$L = 0.5 \text{ in}$$

$$d = 2.5 \text{ in}$$

$$d_o = 6.25 \text{ in}$$

$$d_i = 2 \text{ in}$$

Note: 2.5H7/s6

4.2 Fit Dimensions

$$\delta_F = 0.0021 \text{ in}$$

$$d_{min} = d + \delta_F = 2.5021 \text{ in}$$

$$\Delta_D = 0.0012 \text{ in}$$

$$D_{max} = d + \Delta_D = 2.5012 \text{ in}$$

$$\delta_{min} = d_{min} - D_{max} = 0.0009 \text{ in}$$

4.3 Pressure

$$p = \frac{E\delta_{min}}{2d^3} \left(\frac{(d_o^2 - d^2)(d^2 - d_i^2)}{d_o^2 - d_i^2} \right) = 1813 \text{ psi}$$

4.4 Max Torque

$$T = \frac{\pi}{2} \mu_s p L d = 1424 \text{ in} \cdot \text{lb}$$

Calculations for Fatigue and Contact Stresses on Gears

Part I Setup

1 Material Properties 1050 Steel Q&T @800

$$S_y = 115 \text{ ksi}$$

$$S_{ut} = 158 \text{ ksi}$$

$$H_B = 444$$

$$E = 29.9 \cdot 10^6 \text{ psi}$$

$$\nu = 0.291$$

$$\rho = 0.315969 \text{ lb/in}^3$$

Source for Red Brass density: <https://kg-m3.com/material/red-brass>

2 Centroid and Torque of Minute Hand

2.1 Centroid and Area of Minute Hand without Hole for Shaft

$$C_{tmin} = \frac{82(6 + 2 \cdot 3) \text{ in}^2}{3(6 + 3) \text{ in}} = 36.44 \text{ in}$$

$$A_{tmin} = \frac{82(6 + 3) \text{ in}^2}{2} = 369 \text{ in}^2$$

2.2 Centroid and Area of Hole for Shaft

$$d_{cmin} = 1 \text{ in}$$

$$A_{cmin} = \pi \cdot \left(\frac{d_{cmin}}{2} \right)^2 = 0.7854 \text{ in}^2$$

$$C_{cmin} = 12 \text{ in}$$

$$C_{min} = \frac{C_{tmin}A_{tmin} - C_{cmin}A_{cmin}}{A_{tmin} - A_{cmin}} = 36.497 \text{ in}$$

2.3 Volume of Minute Hand

$$V_{min} = (A_{tmin} - A_{cmin}) 0.5 = 184.108 \text{ in}^3$$

2.4 Torque Generated by Minute Hand

$$T_{min} = (C_{min} - C_{cmin}) \rho V_{min} = 1425.022 \text{ in} \cdot \text{lb}$$

3 Centroid and Torque of Hour Hand

3.1 Centroid and Area of Hour Hand without Hole for Shaft

$$C_{thour} = \frac{50(6 + 2 \cdot 3) \text{ in}^2}{3(6 + 3) \text{ in}} = 22.22 \text{ in}$$

$$A_{thour} = \frac{50(6 + 3)}{2} = 225 \text{ in}^2$$

3.2 Centroid and Area of Hole for Shaft

$$d_{chour} = 2.75 \text{ in}^2$$

$$A_{chour} = \pi \cdot \left(\frac{d_{chour}}{2} \right)^2 = 5.940 \text{ in}^2$$

$$C_{chour} = 10 \text{ in}$$

$$C_{hour} = \frac{C_{thour}A_{thour} - C_{chour}A_{chour}}{A_{thour} - A_{chour}} = 22.554 \text{ in}$$

3.3 Volume of Hour Hand

$$V_{hour} = (A_{thour} - A_{chour}) 0.5 = 109.530 \text{ in}^3$$

3.4 Torque Generated by Minute Hand

$$T_{hour} = (C_{hour} - C_{chour}) \rho V_{hour} = 434.458 \text{ in} \cdot \text{lb}$$

4 Diameter of Gears

4.1 Pitch of Teeth

$$P = 12$$

4.2 Number of Teeth Per Gear

$$N_2 = 20 = 20$$

$$N_3 = 80 = 80$$

$$N_4 = 25 = 25$$

$$N_5 = 75 = 75$$

4.3 Gear Diameter Calculations

$$D_2 = \frac{N_2}{P} = 1.667 \text{ in}$$

$$D_3 = \frac{N_3}{P} = 6.67 \text{ in}$$

$$D_4 = \frac{N_4}{P} = 2.083 \text{ in}$$

$$D_5 = \frac{N_5}{P} = 6.250 \text{ in}$$

Part II

Fatigue, Lewis Equation

5 Forces in Gears

$$W_{t5} = \frac{T_{hour}}{\frac{D_5}{2}} = 139.026 \text{ lb}$$

$$W_{t4} = W_{t5} = 139.026 \text{ lb}$$

$$W_{t3} = \frac{(W_{t4} \cdot \frac{D_4}{2})}{\frac{D_3}{2}} = 43.446 \text{ lb}$$

$$W_{t2} = W_{t3} = 43.446 \text{ lb}$$

6 Torques in Gears

$$T_5 = T_{hour} = 434.458 \text{ in} \cdot \text{lb}$$

$$T_4 = \frac{T_5}{3} = 144.819 \text{ in} \cdot \text{lb}$$

$$T_3 = T_4 = 144.819 \text{ in} \cdot \text{lb}$$

$$T_2 = \frac{T_3}{4} = 36.205 \text{ in} \cdot \text{lb}$$

7 Parameters

7.1 Pressure Angle

$$\phi = 20^\circ$$

7.2 Face Width

$$F = 0.5 \text{ in}$$

7.3 Tooth Width

$$t = \frac{1.561}{P} = 0.130$$

7.4 Tooth Depth

$$h = \frac{2.25}{P} = 0.1875$$

$$L = h = 0.1875$$

8 Gear 2 Fatigue

8.1 Angular Velocity

Note: $1 \frac{rev}{hr} = \frac{\pi}{30} \frac{rad}{min}$

$$\omega_2 = \frac{\pi}{30} = 0.1047 \frac{rad}{min}$$

8.2 Form Factor (Table 14-2)

$$Y_2 = 0.322$$

8.3 Barth Equation

Note: 1/12 is in to feet

$$V_2 = \frac{1}{12} \omega_2 \cdot \frac{D_2}{2} = 0.007272$$

$$K_{v2} = \sqrt{\frac{78 + \sqrt{V_2}}{78}} = 1.000$$

8.4 Stress Concentration Factor

$$r_2 = \frac{0.9025}{P(0.5N_2 + 0.95)} = 0.006868$$

$$K_{f2} = 0.180 + \left(\frac{t}{r_2}\right)^{0.156} \left(\frac{t}{h}\right)^{0.45} = 1.522$$

8.5 Lewis Equation

$$\sigma_2 = \frac{K_{f2} K_{v2} W_{t2} P}{F Y_2} = 4932 \text{ psi}$$

8.6 Endurance Strength

8.6.1 Surface Factor (Machined)

$$a = 2$$

$$b = -0.217$$

$$k_a = a S_{ut}^b = 0.6667$$

8.6.2 Size Factor

$$t_{e2} = \sqrt{\frac{6 L Y_2}{P}} = 0.1737$$

$$d_{e2} = 0.808 \sqrt{F \cdot t_{e2}} = 0.2382$$

$$k_{b2} = \begin{cases} 1 & d_{e2} < 0.3 \\ 0.879 d_{e2}^{-0.102} & 0.3 < d_{e2} < 2 \\ 0.091 d_{e2}^{-0.157} & 2 < d_{e2} < 10 \end{cases} = 1$$

8.6.3 Load Factor (Bending)

$$k_c = 1$$

8.6.4 Temperature Factor (Ambient)

$$k_d = 1$$

8.6.5 Reliability Factor (99%, Table 6-4)

$$k_e = 0.814$$

8.6.6 Miscellaneous Effects Factor (Reversible Gearing)

$$k_f = 1$$

8.6.7 Endurance Strength

$$S_{e2} = k_a k_{b2} k_c k_d k_e k_f \cdot \frac{S_{ut}}{2} = 42.872 \text{ ksi}$$

8.7 Goodman Equation

$$n_{f2} = \frac{S_{e2}}{\frac{\sigma_2}{1000}} = 8.69$$

9 Gear 3 Fatigue

9.1 Angular Velocity

$$\omega_3 = \frac{\omega_2}{4} = 0.02618 \frac{\text{rad}}{\text{min}}$$

9.2 Form Factor (Table 14-2)

$$Y_3 = 0.435$$

9.3 Barth Equation

$$V_3 = V_2 = 0.007272$$
$$K_{v3} = \sqrt{\frac{78 + \sqrt{V_3}}{78}} = 1.001$$

9.4 Stress Concentration Factor

$$r_3 = \frac{0.9025}{P(0.5N_3 + 0.95)} = 0.001837$$
$$K_{f3} = 0.180 + \left(\frac{t}{r_3}\right)^{0.156} \left(\frac{t}{h}\right)^{0.45} = 1.829$$

9.5 Lewis Equation

$$\sigma_3 = \frac{K_{f3} K_{v3} W_{t3} P}{F Y_2} = 5925 \text{ psi}$$

9.6 Endurance Strength

9.6.1 Size Factor

$$t_{e3} = \sqrt{\frac{6 L Y_3}{P}} = 0.2019$$
$$d_{e3} = 0.808 \sqrt{F \cdot t_{e3}} = 0.2568$$
$$k_{b3} = \begin{cases} 1 & d_{e2} < 0.3 \\ 0.879 d_{e2}^{-0.102} & 0.3 < d_{e2} < 2 = 1 \\ 0.091 d_{e2}^{-0.157} & 2 < d_{e2} < 10 \end{cases}$$

9.6.2 Endurance Strength

$$S_{e3} = k_a k_{b3} k_c k_d k_e k_f \cdot \frac{S_{ut}}{2} = 42.87 \text{ ksi}$$

9.7 Goodman Equation

$$n_{f3} = \frac{S_{e3}}{\frac{\sigma_3}{1000}} = 7.235$$

10 Gear 4 Fatigue

10.1 Angular Velocity

$$\omega_4 = \omega_3 = 0.02618 \frac{\text{rad}}{\text{min}}$$

10.2 Form Factor (Table 14-2)

$$Y_4 = 0.337$$

10.3 Barth Equation

$$V_4 = \frac{1}{12} \omega_4 \cdot \frac{D_4}{2} = 0.002273$$

$$K_{v4} = \sqrt{\frac{78 + \sqrt{V_4}}{78}} = 1.000$$

Really don't need this.
It is only for
high speed machines.

10.4 Stress Concentration Factor

$$r_4 = \frac{0.9025}{P(0.5N_4 + 0.95)} = 0.005592$$

$$K_{f4} = 0.180 + \left(\frac{t}{r_4}\right)^{0.156} \left(\frac{t}{h}\right)^{0.45} = 1.566$$

10.5 Lewis Equation

$$\sigma_4 = \frac{K_{f4} K_{v4} W_{t4} P}{F Y_4} = 15509 \text{ psi}$$

10.6 Endurance Strength

10.6.1 Size Factor

$$t_{e4} = \sqrt{\frac{6 L Y_4}{P}} = 0.1777$$

$$d_{e4} = 0.808 \sqrt{F \cdot t_{e4}} = 0.2409$$

$$k_{b4} = \begin{cases} 1 & d_{e2} < 0.3 \\ 0.879 d_{e2}^{-0.102} & 0.3 < d_{e2} < 2 = 1 \\ 0.091 d_{e2}^{-0.157} & 2 < d_{e2} < 10 \end{cases}$$

10.6.2 Endurance Strength

$$S_{e4} = k_a k_{b4} k_c k_d k_e k_f \cdot \frac{S_{ut}}{2} = 42.87 \text{ ksi}$$

10.7 Goodman Equation

$$n_{f4} = \frac{S_{e4}}{\frac{\sigma_4}{1000}} = 2.764$$

11 Gear 5 Fatigue

11.1 Angular Velocity

$$\omega_5 = \frac{\omega_4}{3} = 0.008727 \frac{rad}{min}$$

11.2 Form Factor (Table 14-2)

$$Y_5 = 0.435 = 0.435$$

11.3 Barth Equation

$$V_5 = V_4 = 0.002273$$
$$K_{v5} = \sqrt{\frac{78 + \sqrt{V_5}}{78}} = 1.000$$

11.4 Stress Concentration Factor

$$r_5 = \frac{0.9025}{P(0.5N_5 + 0.95)} = 0.001956$$
$$K_{f5} = 0.180 + \left(\frac{t}{r_5}\right)^{0.156} \left(\frac{t}{h}\right)^{0.45} = 1.813$$

11.5 Lewis Equation

$$\sigma_5 = \frac{K_{f5}K_{v5}W_{t5}P}{FY_5} = 13900 \text{ psi}$$

11.6 Endurance Strength

11.7 Size Factor

$$t_{e5} = \sqrt{\frac{6LY_5}{P}} = 0.2019$$
$$d_{e5} = 0.808\sqrt{F \cdot t_{e5}} = 0.2568$$
$$k_{b5} = \begin{cases} 1 & d_{e2} < 0.3 \\ 0.879d_{e2}^{-0.102} & 0.3 < d_{e2} < 2 \\ 0.091d_{e2}^{-0.157} & 2 < d_{e2} < 10 \end{cases} = 1$$

11.7.1 Endurance Strength

$$S_{e5} = k_a k_{b5} k_c k_d k_e k_f \cdot \frac{S_{ut}}{2} = 42.87 \text{ ksi}$$

11.8 Goodman Equation

$$n_{f5} = \frac{S_{e5}}{\frac{\sigma_5}{1000}} = 3.082$$

Part III

Contact Stress

12 Contact Strength

Note: Grade 1

$$S_c = 322H_B + 29100 = 172068 \text{ psi}$$

13 Stress Between Gears 2 and 3

13.1 Elastic Coefficient

$$C_p = \left(2\pi \cdot \frac{1 - \nu^2}{E} \right)^{-\frac{1}{2}} = 2280.1$$

13.2 Contact Stress

$$\sigma_{23} = -C_p \left(\frac{K_{v2}W_{t2}}{F \cos(\phi)} \left(\frac{1}{\frac{D_2}{2}} + \frac{1}{\frac{D_3}{2}} \right) \right)^{\frac{1}{2}} = -26860 \text{ psi}$$

13.3 Safety Factor

$$n_{23} = \frac{S_c^2}{\sigma_{23}^2} = 41.04$$

14 Stress Between Gears 4 and 5

14.1 Contact Stress

$$\sigma_{45} = -C_p \left(\frac{K_{v4}W_{t4}}{F \cos(\phi)} \left(\frac{1}{\frac{D_4}{2}} + \frac{1}{\frac{D_5}{2}} \right) \right)^{\frac{1}{2}} = -44381 \text{ psi}$$

14.2 Safety Factor

$$n_{23} = \frac{S_c^2}{\sigma_{45}^2} = 15.03$$

Other Considerations

The assembly of each shaft and gear was thought of to ensure that the bearings would butt against each spacer, under the assumption that housings could retain the bearings through either a pressfit or flange. This would retain the bearings between the housing plates, as well as the clock face, to prevent axial motion on any shaft.

Press fits were used on each gear that were H7/s6, a tight interference fit that should retain each gear well. Gears were butted against steps in the shaft to assist in assembly, with chamfers to prevent interference with the steps' fillets.

Summary

All of the specifications of the design were met with our approach. Every piece was calculated to ensure a functionality of infinite life, including a 99.7% reliability rating for the bearings. Every safety factor was targeted to exceed a minimum of 1.5, with critical safety factors generally tuned to fall in the 1.5-2.0 range if possible.

This geartrain was approached with the intent to make it with as small footprint as possible, leading to greater loadcases that push the boundaries of the geometry and require increasingly stronger materials. If the geartrain was instead approached with the intent to make it functional within the footprint of the clock face, roughly twice the length of the minute hand, then the loads would have been lesser, allowing a cheaper and weaker material to be used that could have had internal pockets for the gears. This could potentially save cost in the final design. Under a lesser load, other methods than pressfits and spacers that are simpler and require less precision, like set screws or keying, may have also been better choices for how we retained our components on the shaft. We additionally found that we were pushing the limits of our bearings, having to replace two under the load case we determined.